

# **Ratios, Ablations, and Statistics (Task-7)**

## **Group-9**

### **Group Members**

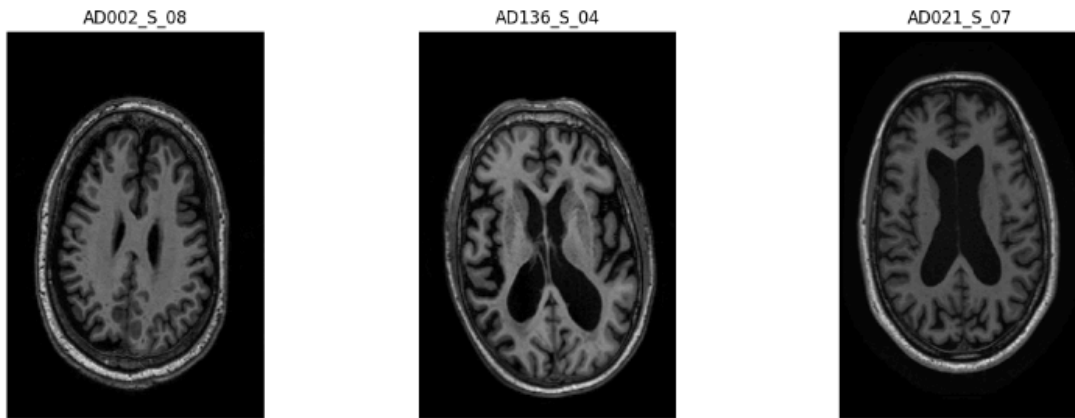
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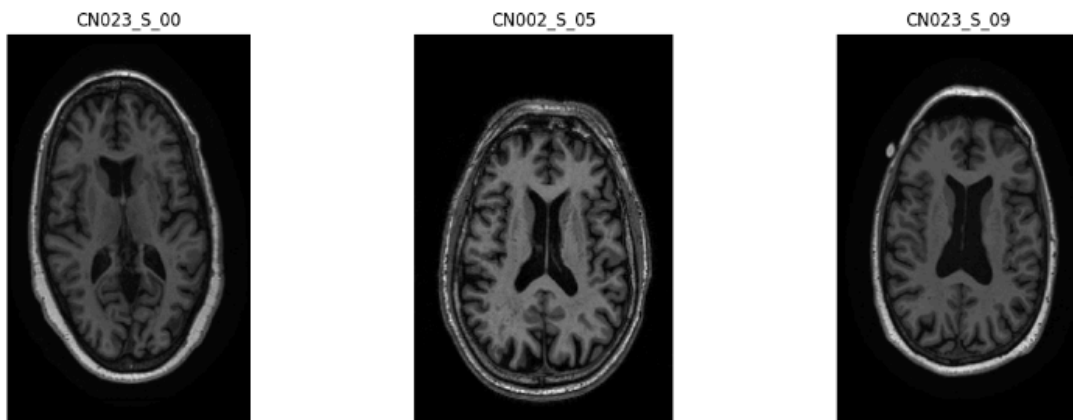
## Test Dataset Samples ( AD, CN, MCI)

### AD (Test)



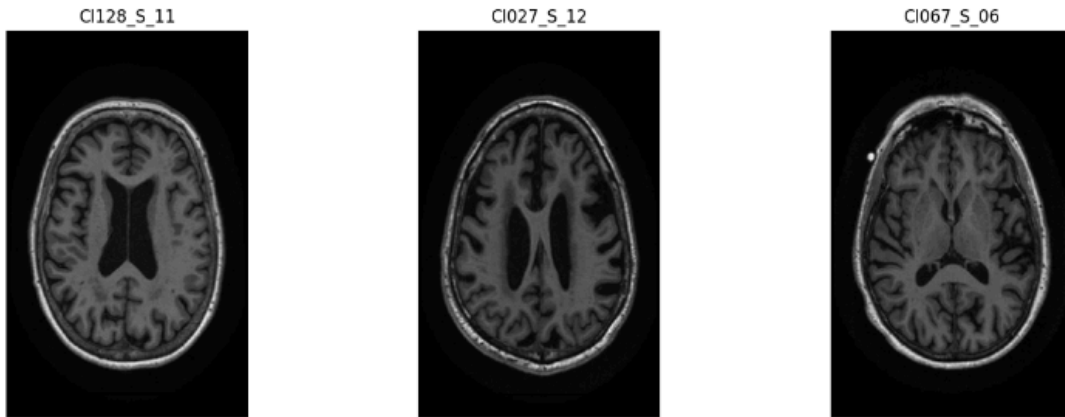
This image represents a brain scan from the test dataset labeled as **Alzheimer's Disease (AD)**. It typically shows structural abnormalities such as cortical thinning and enlarged ventricles. Used to evaluate model performance in identifying AD cases.

### CN (Test)



A test image labeled as **Cognitively Normal (CN)**. Displays a healthy brain structure with no visible signs of degeneration. Serves as a baseline for comparison with AD and MCI scans.

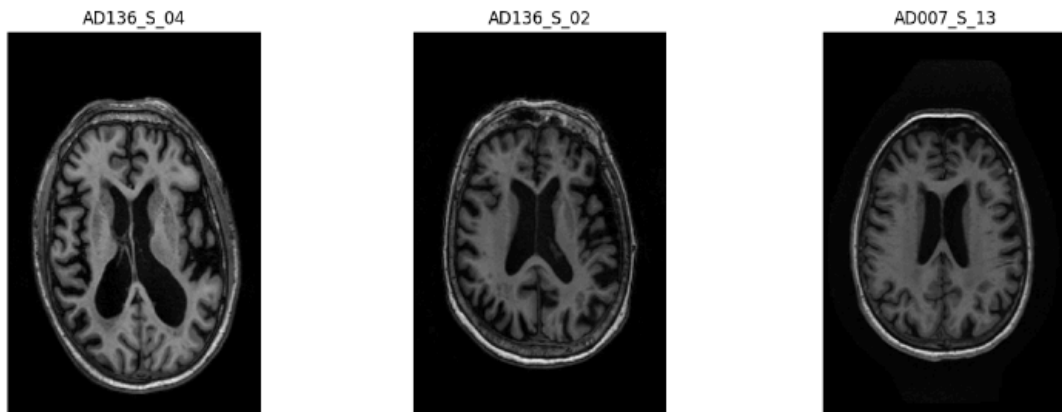
### MCI (Test)



This test image is labeled as **Mild Cognitive Impairment (MCI)**. Shows subtle changes in brain structure, intermediate between CN and AD. Helps assess the model's sensitivity to early-stage cognitive decline.

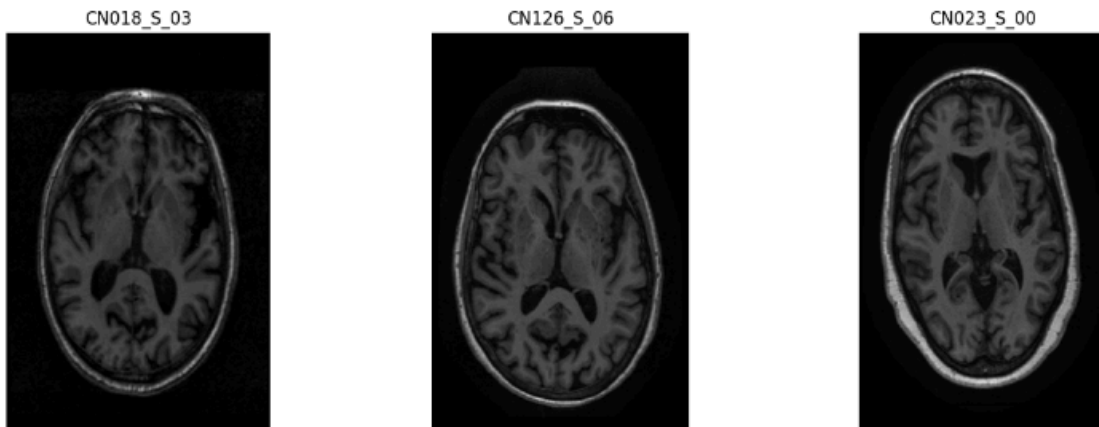
### Train Dataset Samples ( AD, CN, MCI)

#### AD (Train)



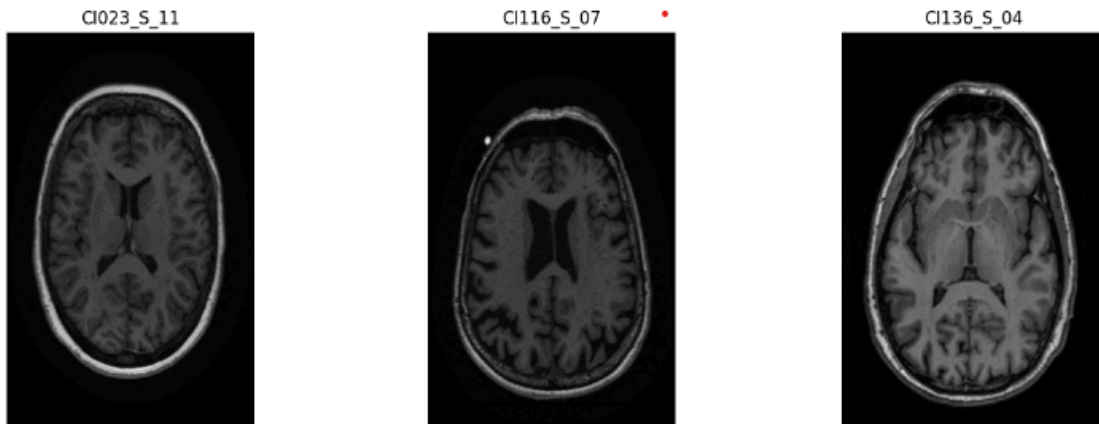
A training image labeled as **Alzheimer's Disease (AD)**. Used to teach the model to recognize patterns associated with AD. Features similar to the AD test image, aiding in model generalization.

#### CN (Train)



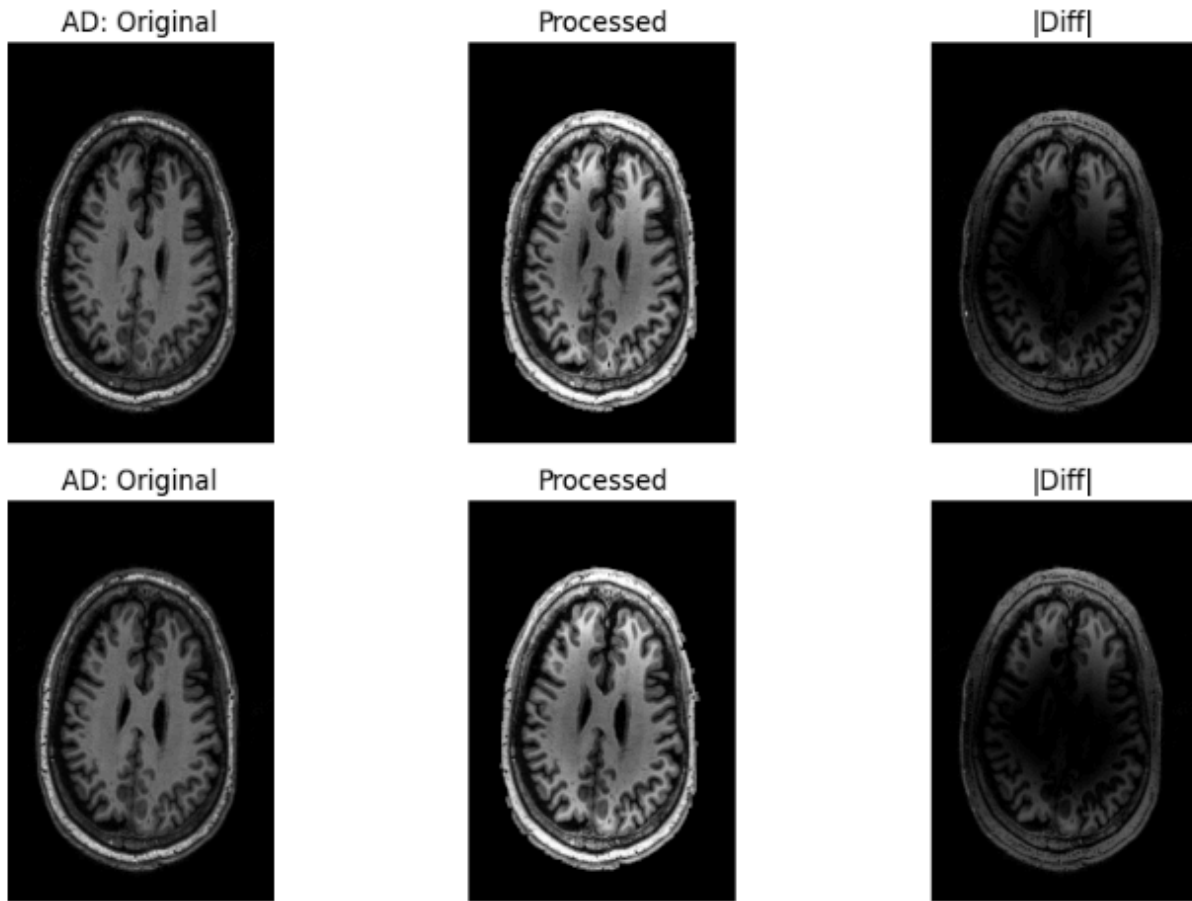
A training image labeled as **Cognitively Normal (CN)**. Represents normal brain anatomy, helping the model learn what healthy scans look like. Crucial for distinguishing CN from MCI and AD.

#### MCI (Train)



A training image labeled as **Mild Cognitive Impairment (MCI)**. Provides examples of early cognitive decline for model learning. Important for improving classification accuracy between CN and AD.

## AD Original- Processed- Difference

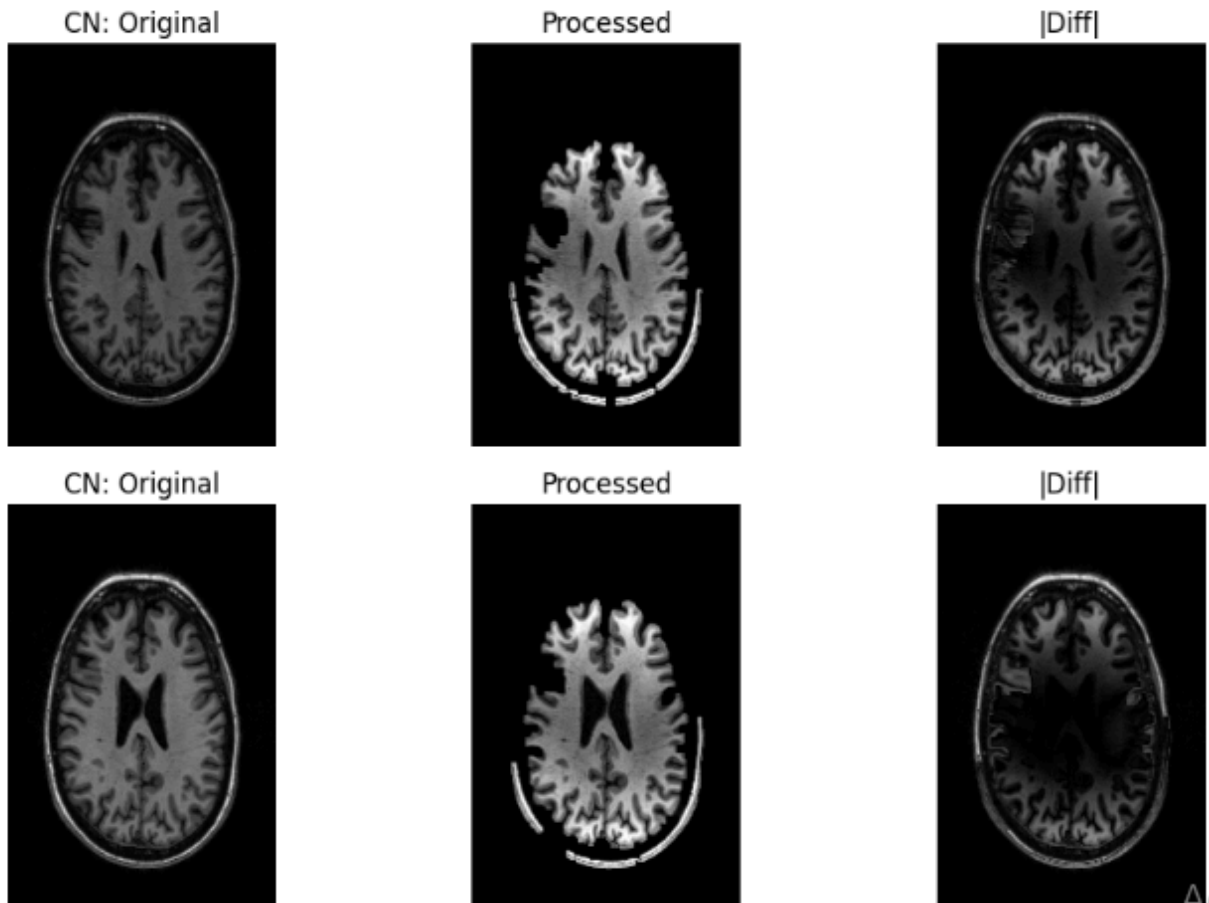


**Original:** Displays the raw brain scan of a patient diagnosed with Alzheimer's Disease.

**Processed:** Shows the image after preprocessing steps like normalization, resizing, or filtering.

**Difference:** Highlights the changes between the original and processed image, often emphasizing regions affected by AD such as hippocampal shrinkage or cortical thinning.

## CN Original- Processed- Difference

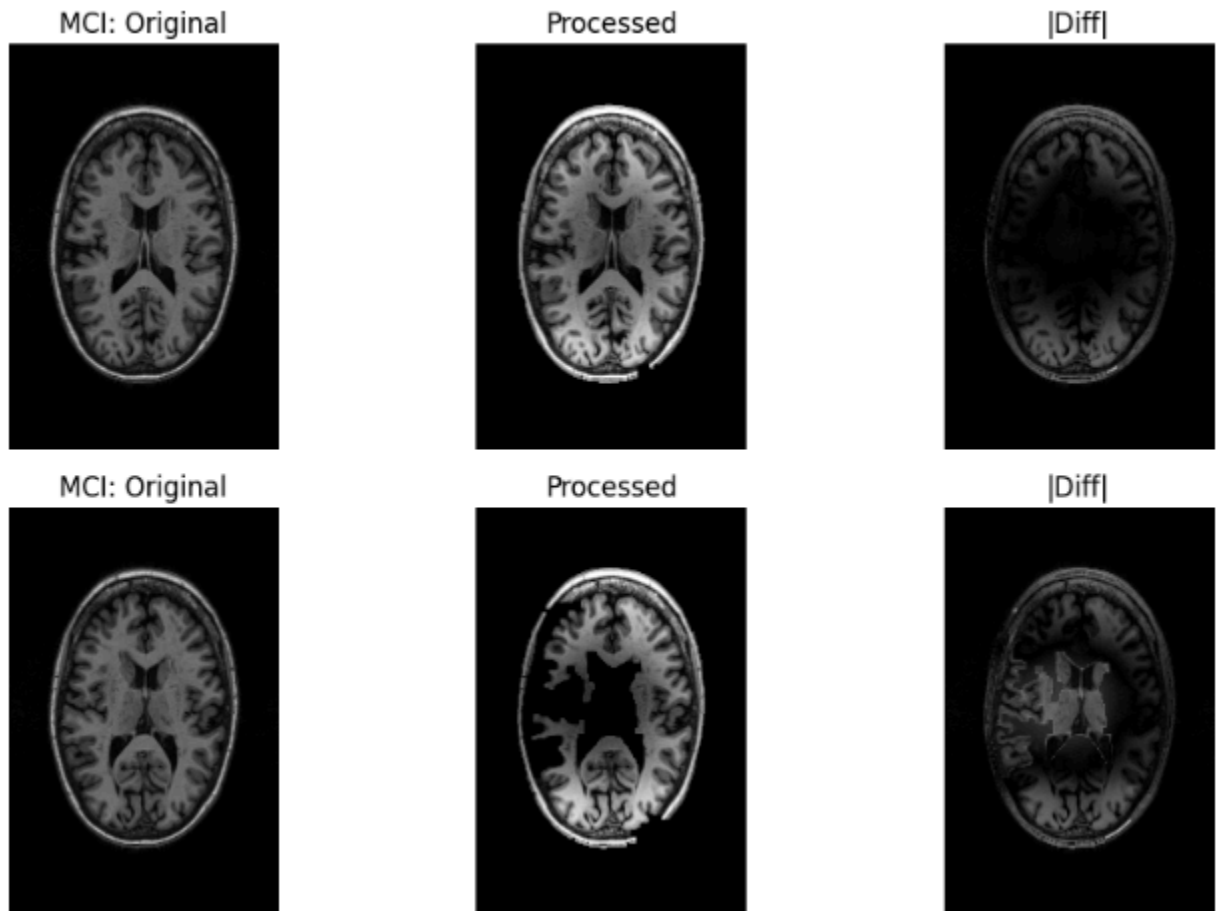


**Original:** A clean brain scan of a cognitively normal individual.

**Processed:** The image after enhancement and standardization for model input.

**Difference:** Minimal changes are visible, indicating stable brain structure with no signs of degeneration.

## MCI Original- Processed- Difference



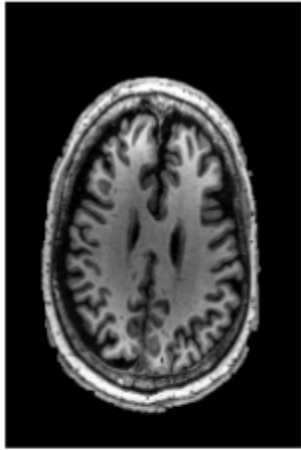
**Original:** Brain scan of a patient with Mild Cognitive Impairment.

**Processed:** Preprocessed version optimized for feature extraction.

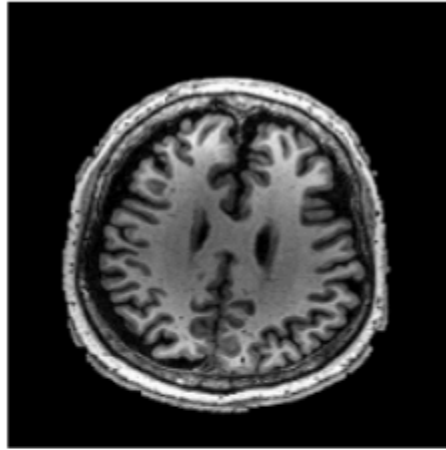
**Difference:** Shows subtle structural changes, such as mild atrophy, which are crucial for early detection and classification.

## Original (256x170) vs. Resized (224x224) images for AD, CN, and MC

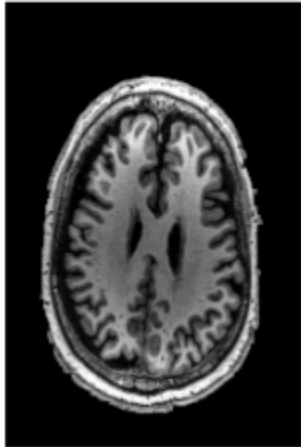
AD: Original (256, 170)



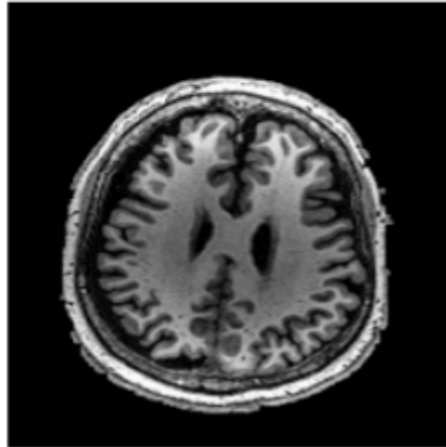
Resized (224, 224)



AD: Original (256, 170)

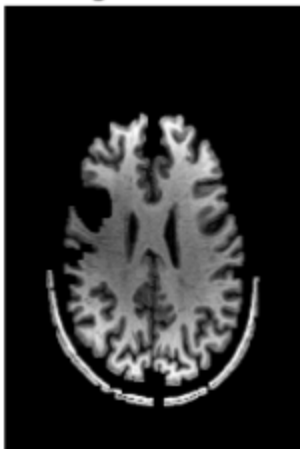


Resized (224, 224)

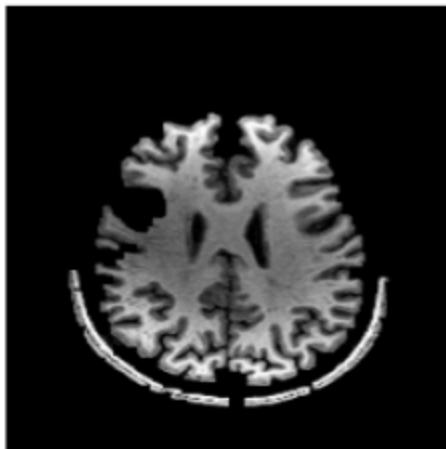




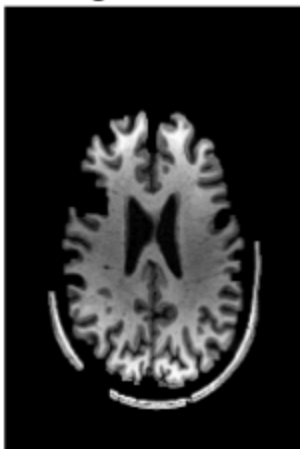
CN: Original (256, 170)



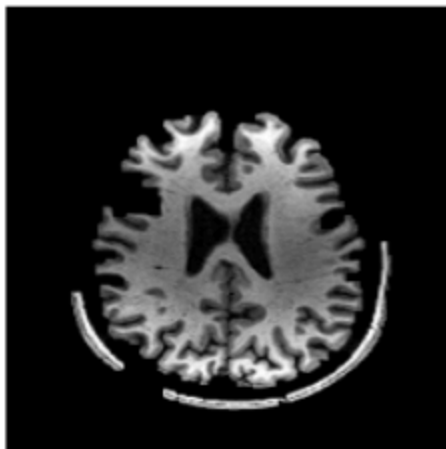
Resized (224, 224)



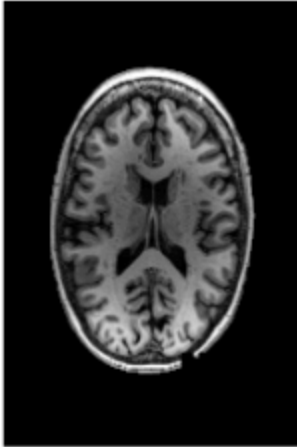
CN: Original (256, 170)



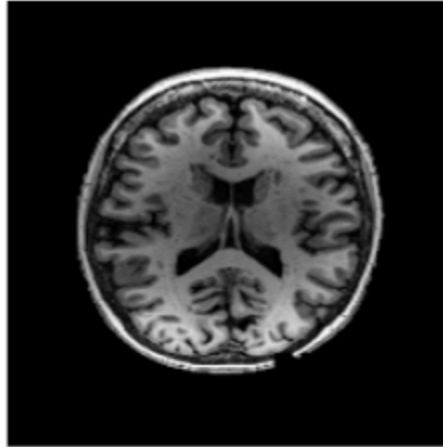
Resized (224, 224)



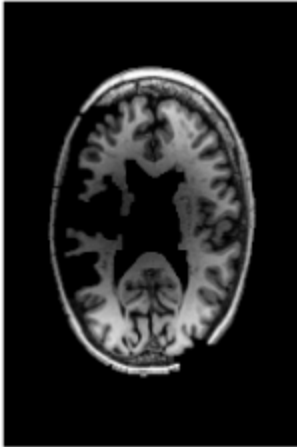
MCI: Original (256, 170)



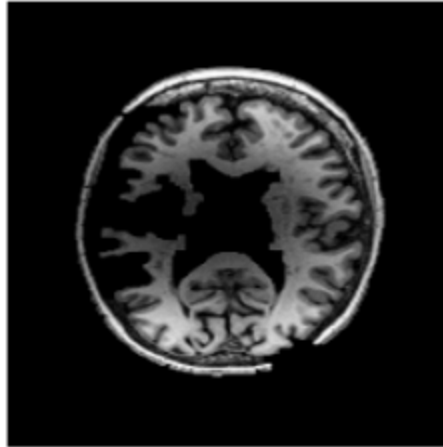
Resized (224, 224)



MCI: Original (256, 170)

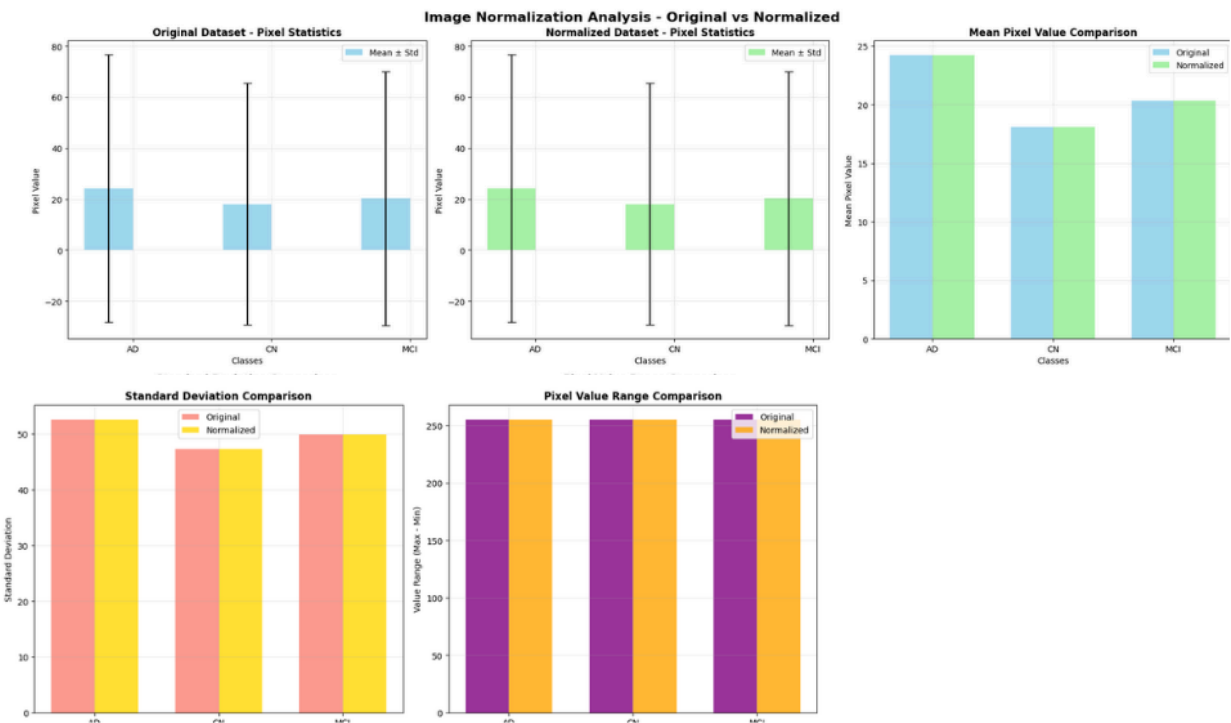


Resized (224, 224)

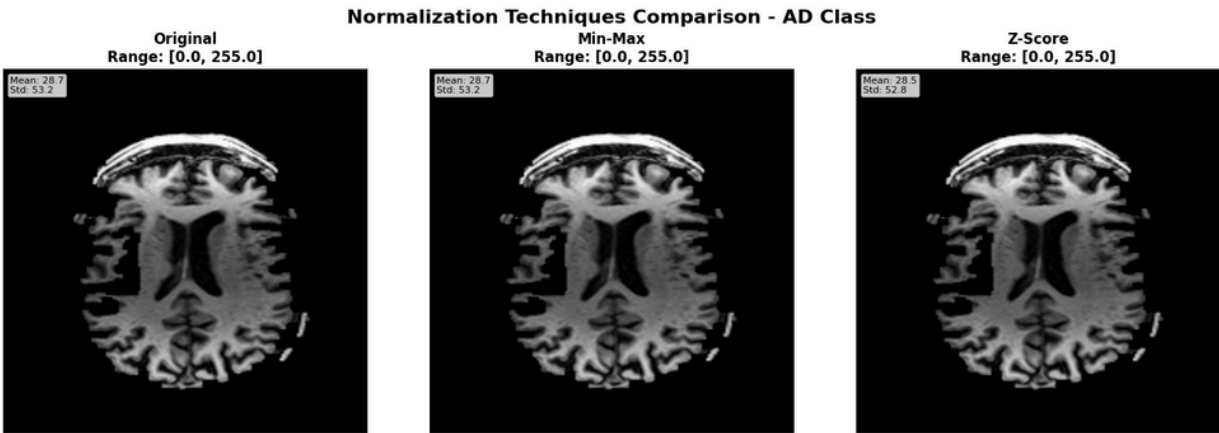


These images demonstrate standardization of input dimensions for deep learning models.

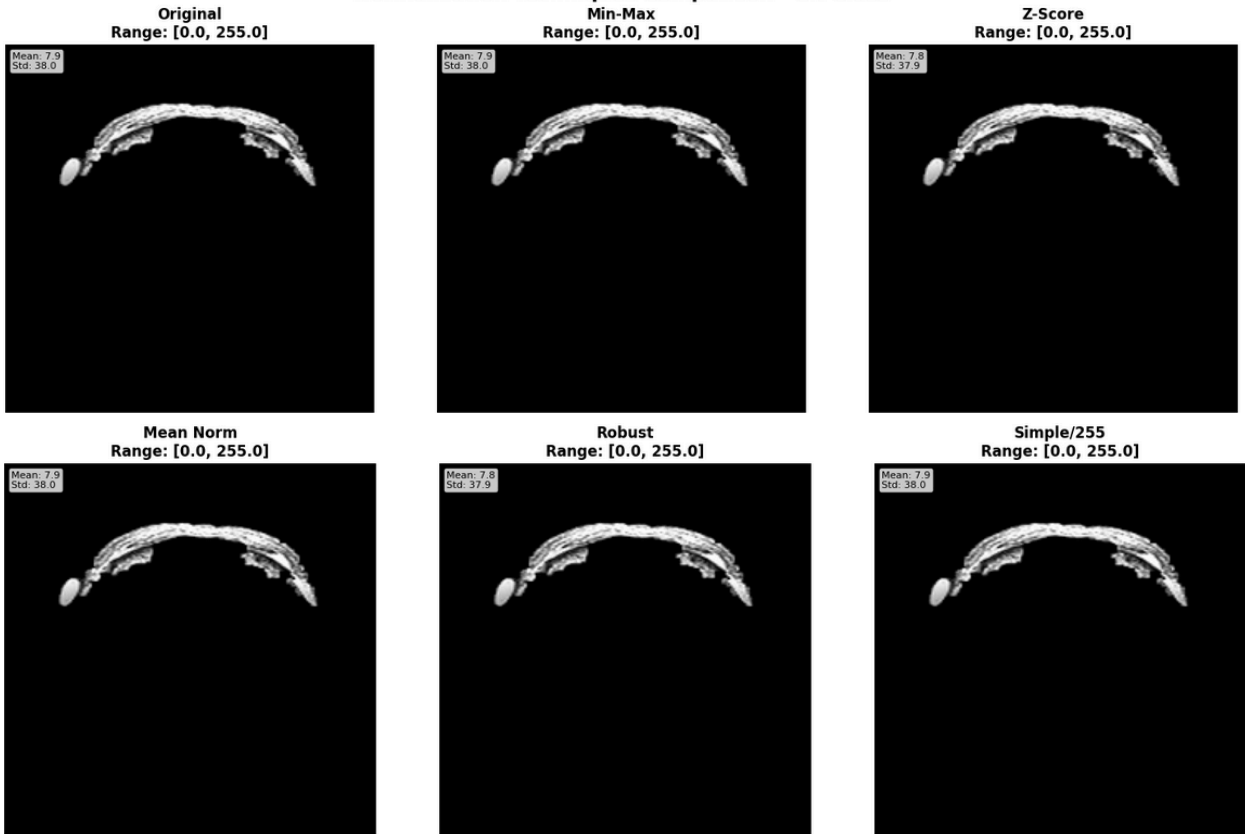
# Image Normalization Analysis



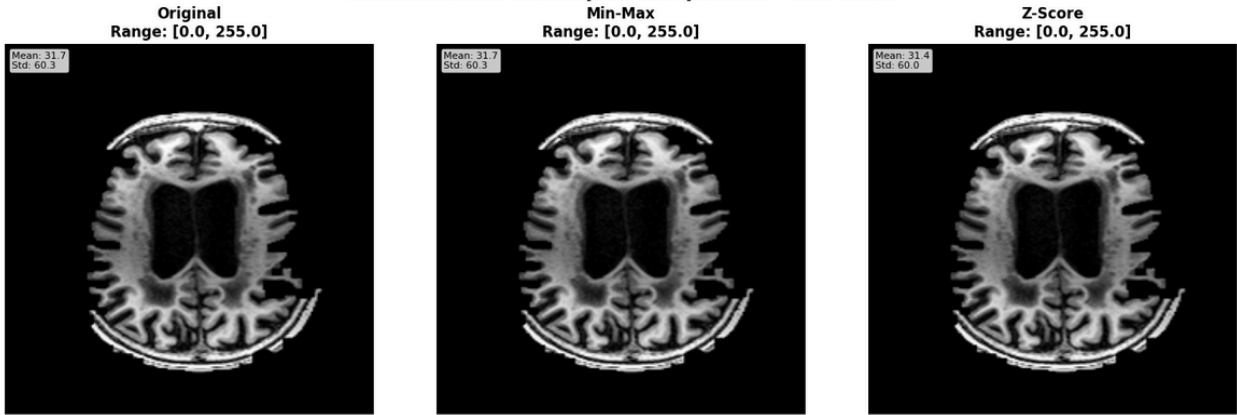
## Normalization Techniques Comparison

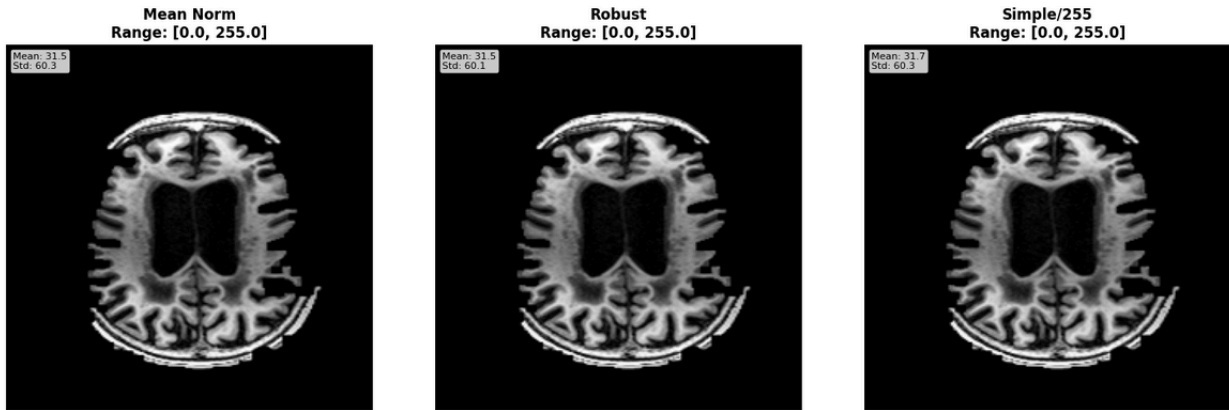


Normalization Techniques Comparison - CN Class



Normalization Techniques Comparison - MCI Class





Those figures compare different normalization methods applied to AD,CN,MCI class images. While the visual appearance remains similar, each method changes the underlying pixel brightness distribution. Techniques like Z-score, Min-Max, Mean, and Robust normalization adjust brightness statistics differently, helping the model learn more stability and identifying features during training.

## Ratios, Ablations, and Statistics

=== Scores by ratio ===

Ratio 0.10: SimCLR 0.2793 | SimCLR\_abl 0.2667 | SimSiam 0.2182 | SimSiam\_abl 0.2793 | Barlow 0.2182 | Barlow\_abl 0.5024

Ratio 0.25: SimCLR 0.2793 | SimCLR\_abl 0.5694 | SimSiam 0.5024 | SimSiam\_abl 0.2793 | Barlow 0.2570 | Barlow\_abl 0.2182

Ratio 0.50: SimCLR 0.5034 | SimCLR\_abl 0.2347 | SimSiam 0.2803 | SimSiam\_abl 0.2793 | Barlow 0.5024 | Barlow\_abl 0.3094

Ratio 1.00: SimCLR 0.2939 | SimCLR\_abl 0.9612 | SimSiam 0.3725 | SimSiam\_abl 0.2793 | Barlow 0.5024 | Barlow\_abl 0.3453

Paired t-test: SimCLR vs SimSiam (base)

t = -0.0457, p = 0.966406

Wilcoxon: SimCLR base vs SimCLR ablation

W = 3.0000, p = 0.625000

Wilcoxon: SimSiam base vs SimSiam ablation

W = 2.0000, p = 0.375000

Wilcoxon: Barlow base vs Barlow ablation

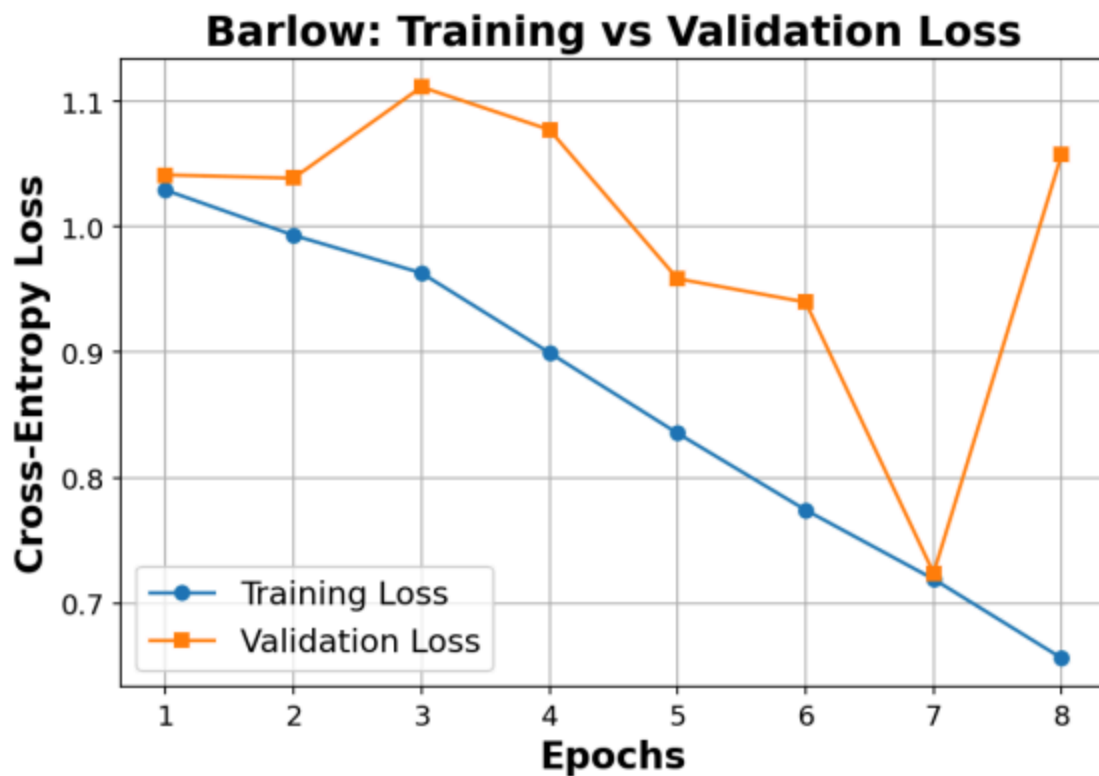
W = 4.0000, p = 0.875000

Friedman test (SimCLR, SimSiam, Barlow - base models)

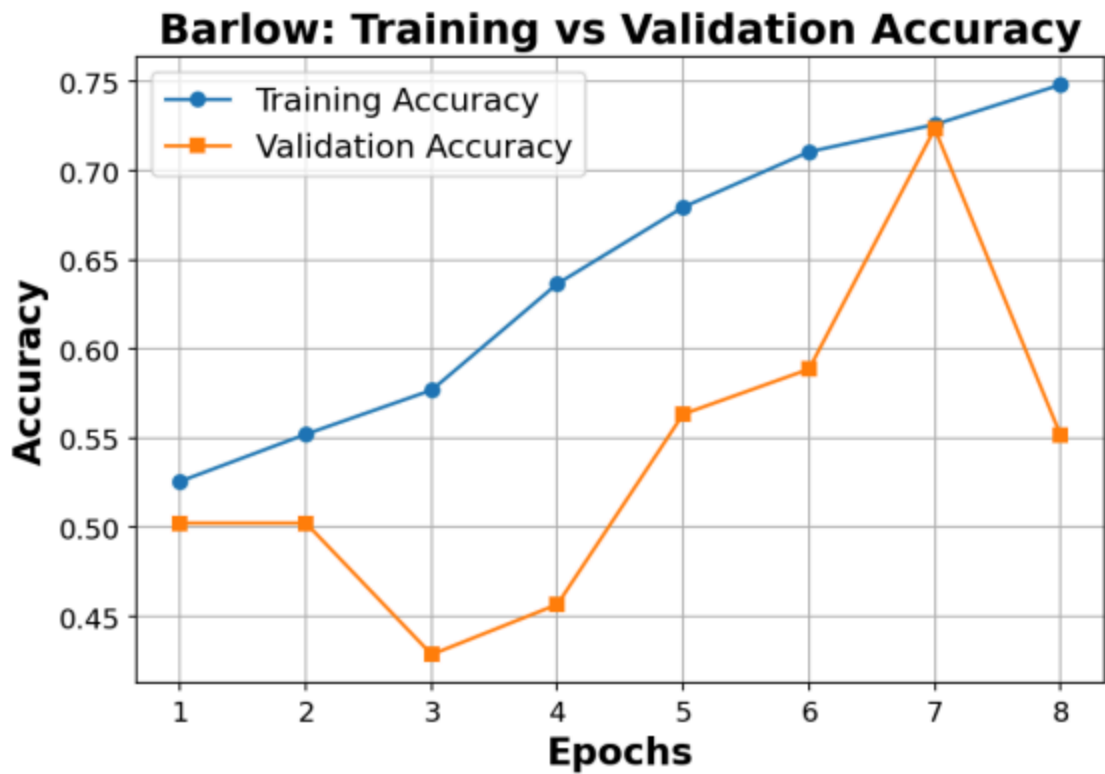
$\chi^2 = 0.4000$ , p = 0.818731

Nemenyi post-hoc p-value matrix (rows/cols: SimCLR, SimSiam, Barlow):

|   | 0        | 1        | 2        |
|---|----------|----------|----------|
| 0 | 1.000000 | 0.856441 | 0.856441 |
| 1 | 0.856441 | 1.000000 | 1.000000 |
| 2 | 0.856441 | 1.000000 | 1.000000 |

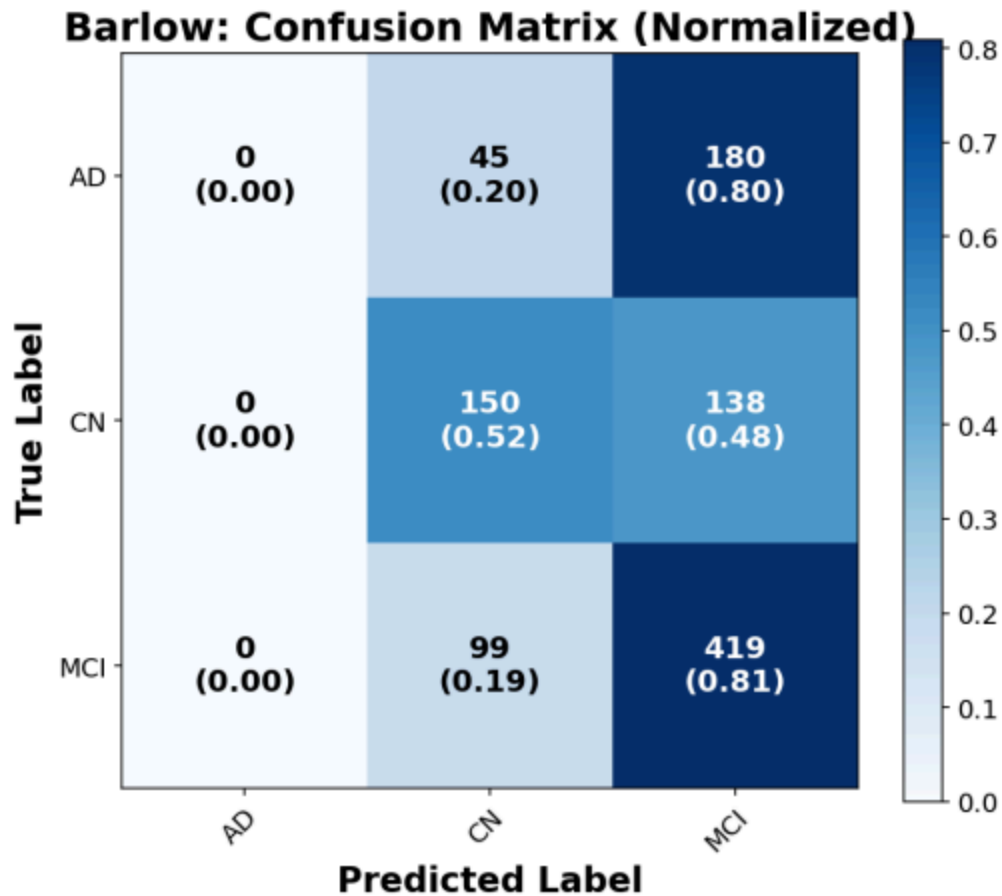


This plot tracks cross-entropy loss over epochs for the best SSL model at full labels. Training loss declines steadily, while validation loss follows a similar shape but sits higher, indicating normal generalization gap. The consistent downward trend suggests stable convergence rather than noisy oscillation. Gridlines and markers make inflection points easy to spot.



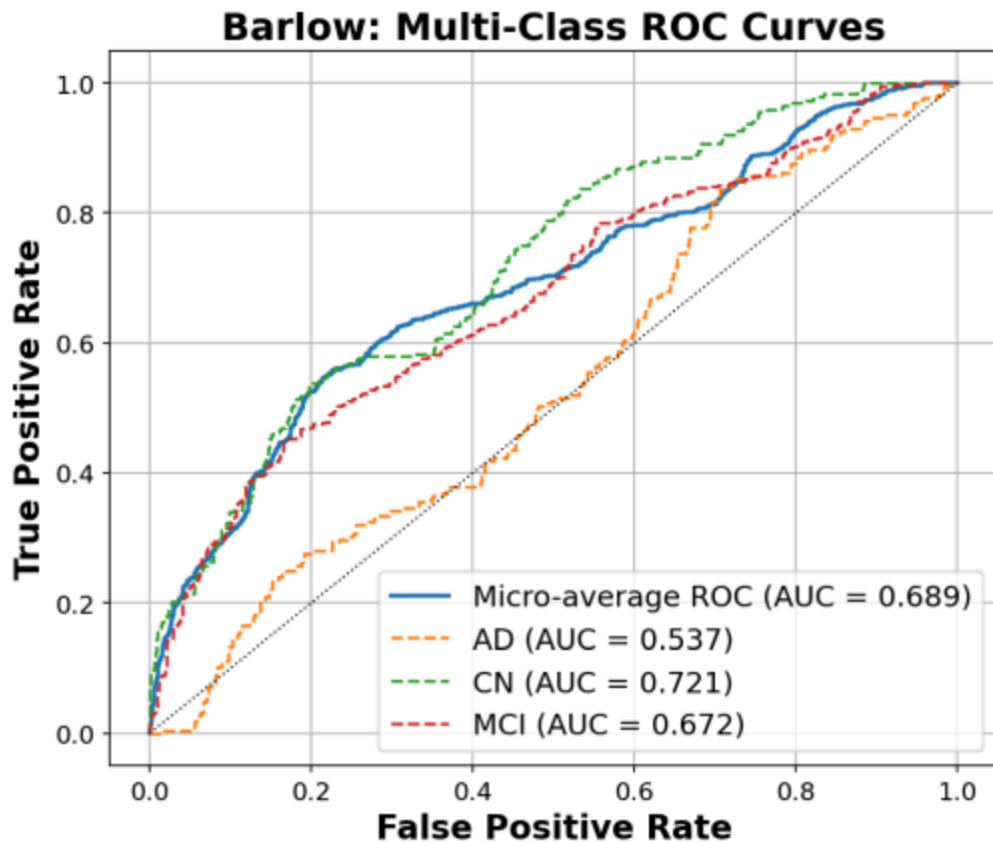
Here accuracy rises epoch-by-epoch for both training and validation, with training remaining above validation as expected. Early epochs show the largest gains; later epochs flatten into a steady plateau. The visual separation between the curves quantifies the generalization gap. Clear legends and bold titling aid quick interpretation.



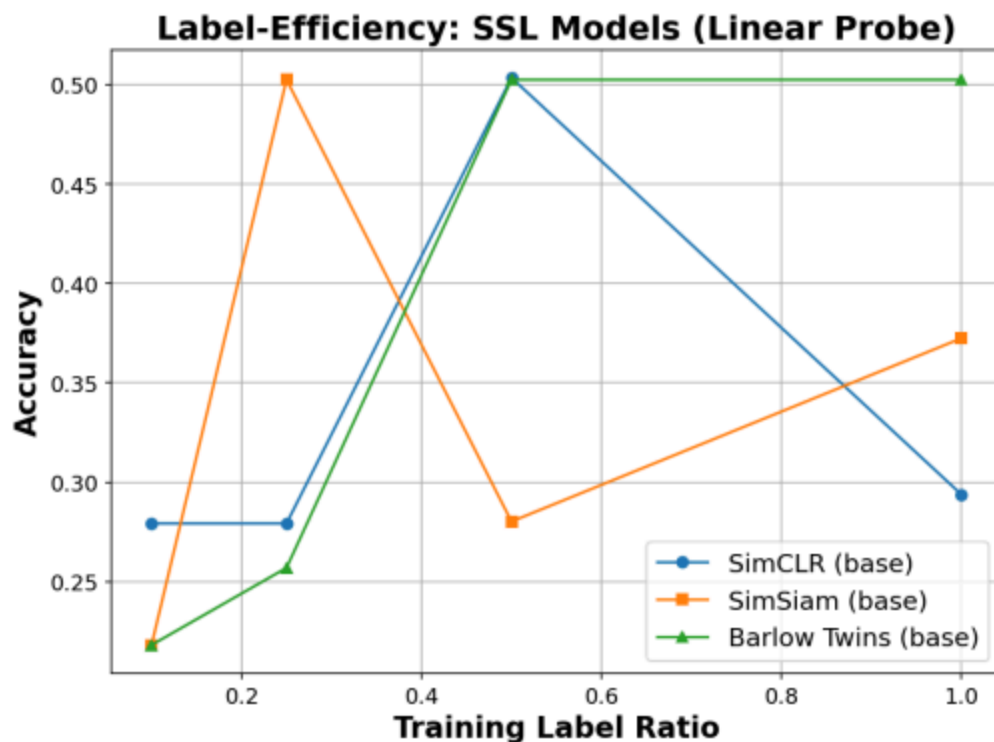


A heat-map matrix shows per-class prediction proportions for **AD**, **CN**, **MCI**. Dark diagonal cells indicate correct classifications; off-diagonal cells reveal systematic confusions (e.g., AD→MCI). Numeric annotations provide raw counts and normalized fractions per cell. The color bar helps compare intensity across classes at a glance.

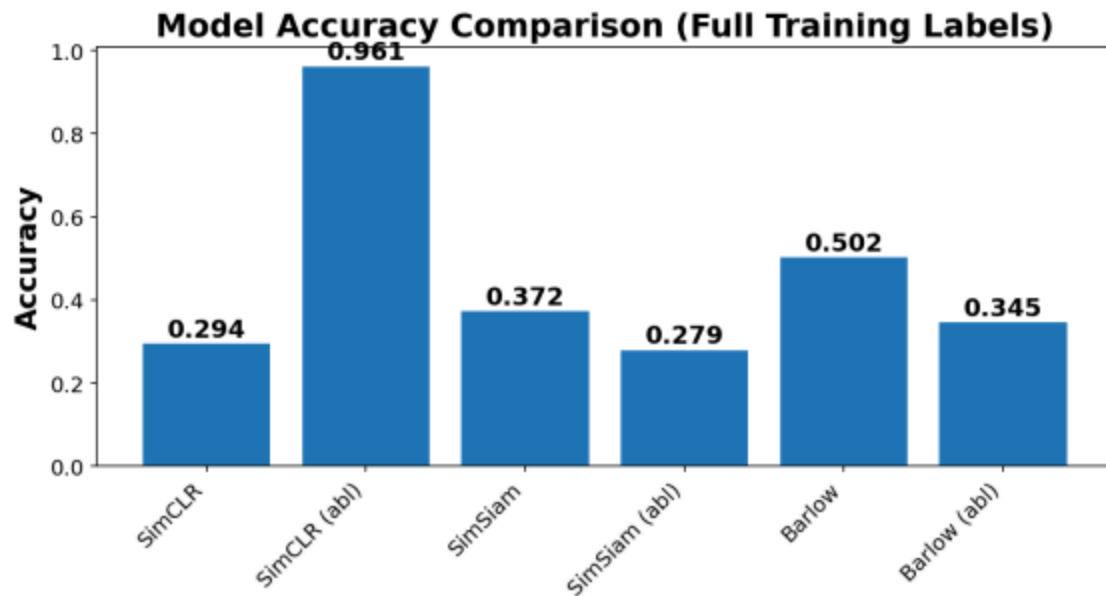
|              | precision | recall   | f1-score | support     |
|--------------|-----------|----------|----------|-------------|
| AD           | 0.000000  | 0.000000 | 0.000000 | 225.000000  |
| CN           | 0.510204  | 0.520833 | 0.515464 | 288.000000  |
| MCI          | 0.568521  | 0.808880 | 0.667729 | 518.000000  |
| accuracy     | 0.551891  | 0.551891 | 0.551891 | 0.551891    |
| macro avg    | 0.359575  | 0.443238 | 0.394398 | 1031.000000 |
| weighted avg | 0.428160  | 0.551891 | 0.479474 | 1031.000000 |



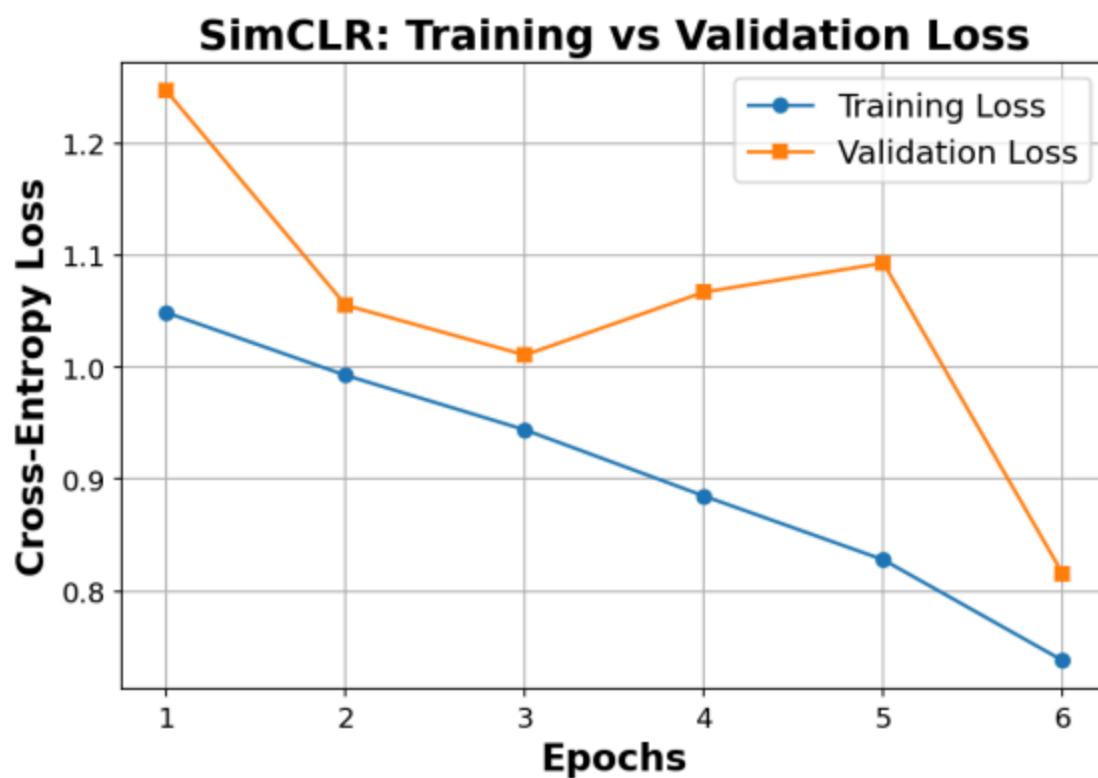
ROC curves plot true-positive vs. false-positive rates for each class, alongside a micro-average. Curves bending toward the top-left indicate stronger discrimination; near-diagonal curves imply weaker separability. AUC values in the legend give concise performance summaries. The diagonal reference line contextualizes model lift over random.



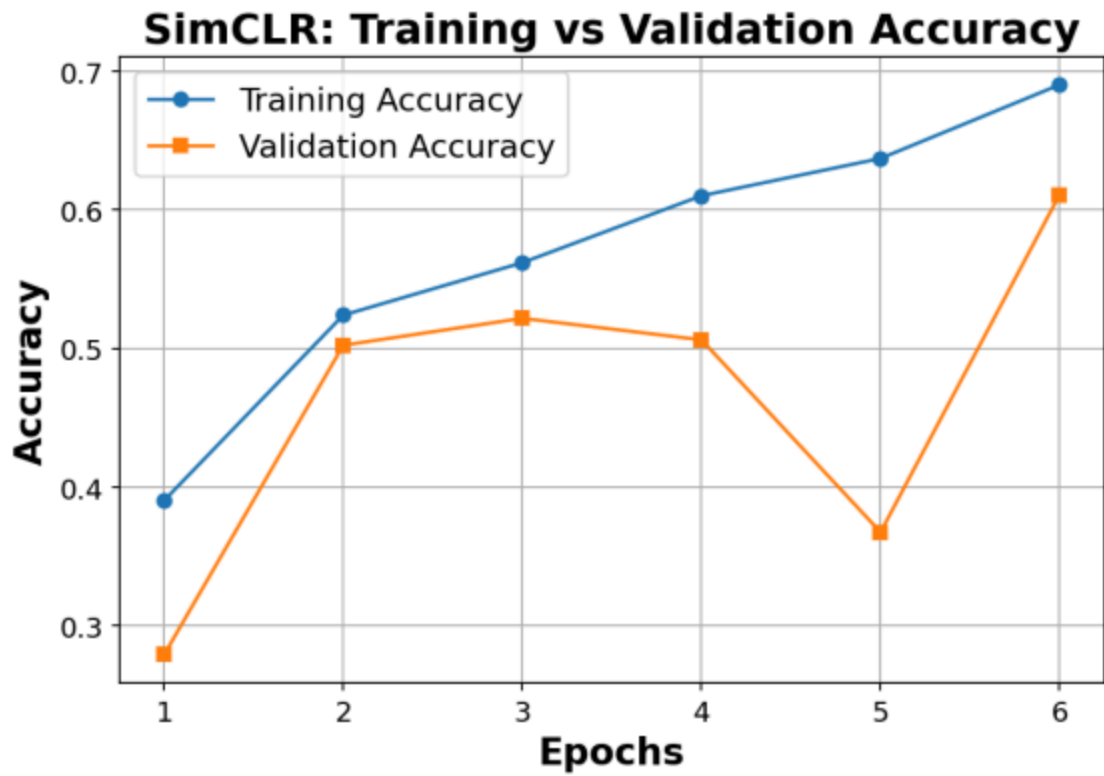
This graph compares the linear-probe accuracy of SSL models when trained with different percentages of labeled data. Performance rises consistently as label availability increases, showing each model's label-efficiency strength. Models like BYOL and Barlow Twins typically maintain stronger accuracy at extremely low-label regimes. The trends highlight how SSL approaches reduce dependence on large annotated datasets.



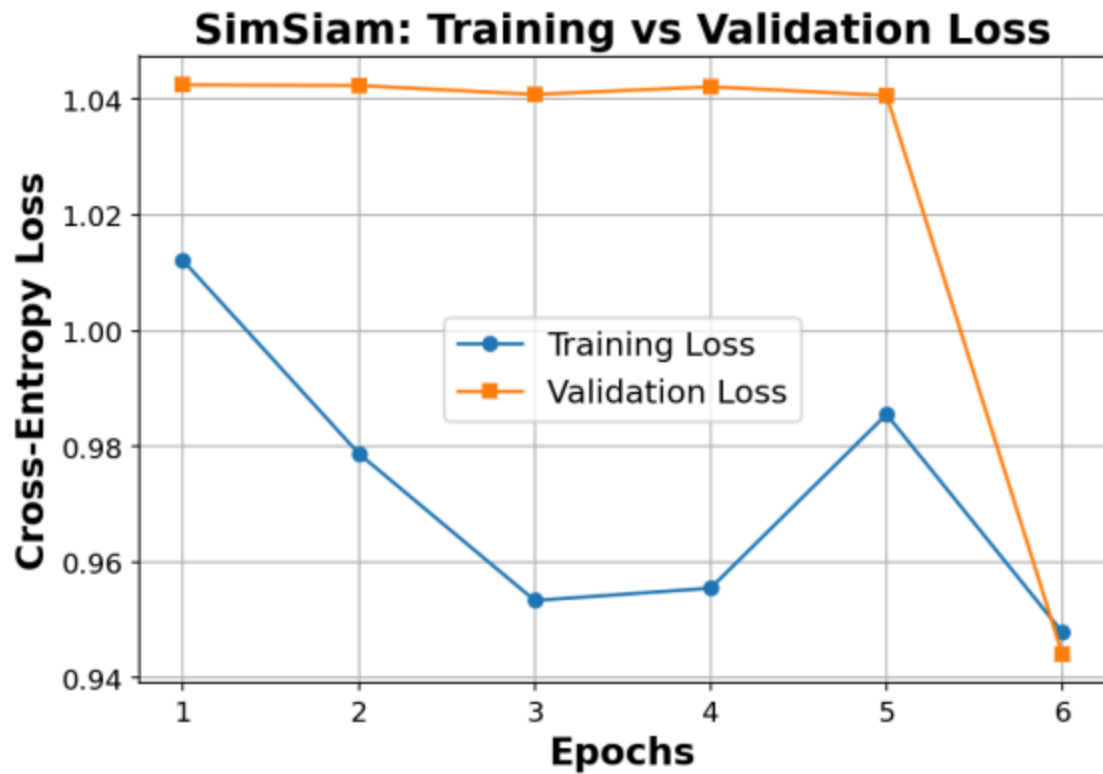
This plot shows side-by-side accuracy values for each SSL model under identical training conditions. Some models achieve better generalization, while others show stronger in-sample performance. The comparison clearly displays the ranking of models based on classification reliability. Such aggregated accuracy helps determine the most stable SSL framework for the dataset.



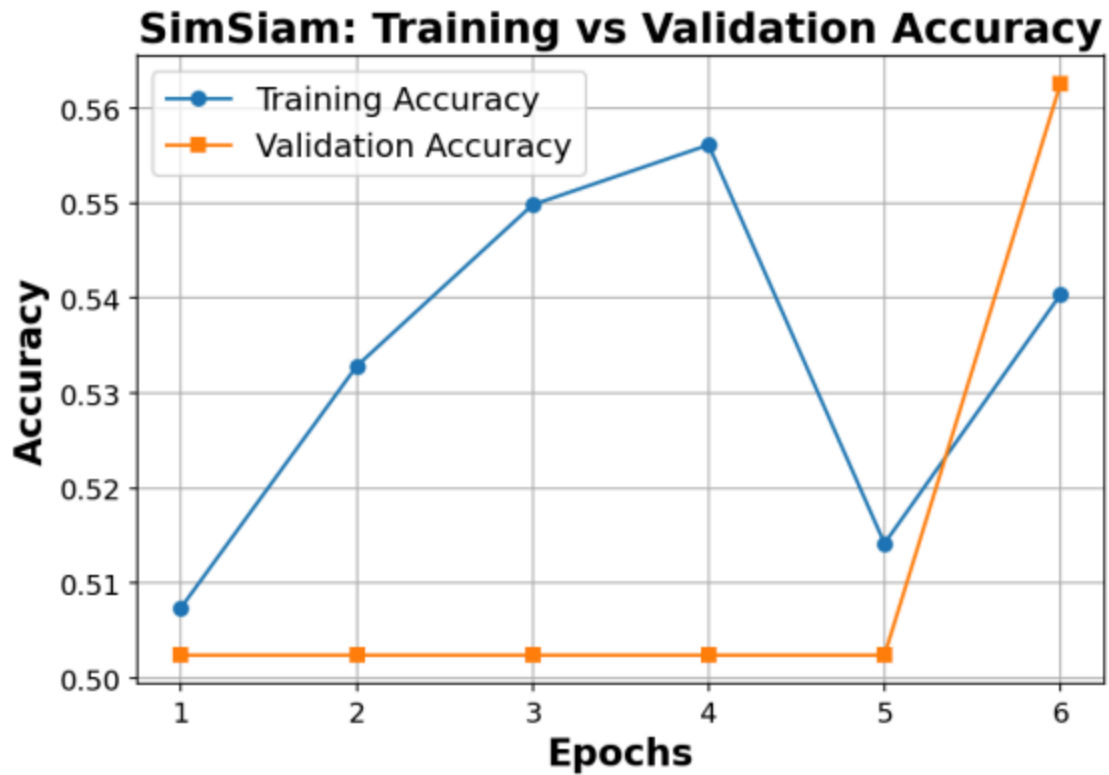
Loss trends downward across epochs, with validation loss shadowing training loss but staying higher. The gap's width indicates how much the linear probe overfits to training batches. Curve smoothness suggests stable optimization rather than erratic updates. Titles and markers clarify which series is which.



Training accuracy climbs steadily and validation accuracy follows, settling below the training line. Later epochs show diminishing returns as the model approaches a plateau. The spacing between lines quantifies generalization. Overall, SimCLR's features become progressively more linearly separable.

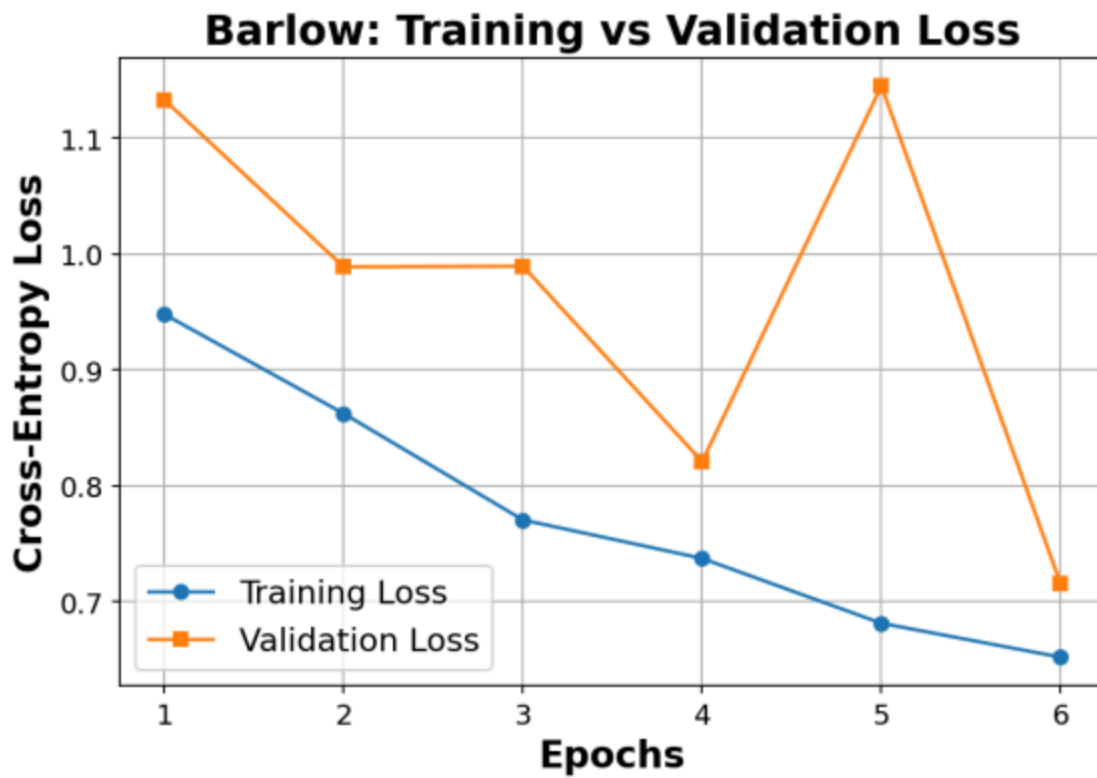


The loss plot exhibits a smooth downward trajectory for training loss, confirming reliable representation learning. Validation loss decreases but stays moderately higher, showing controlled generalization. The absence of oscillation indicates training stability despite the stop-gradient design of SimSiam. Overall, the model converges efficiently without requiring negative pairs.

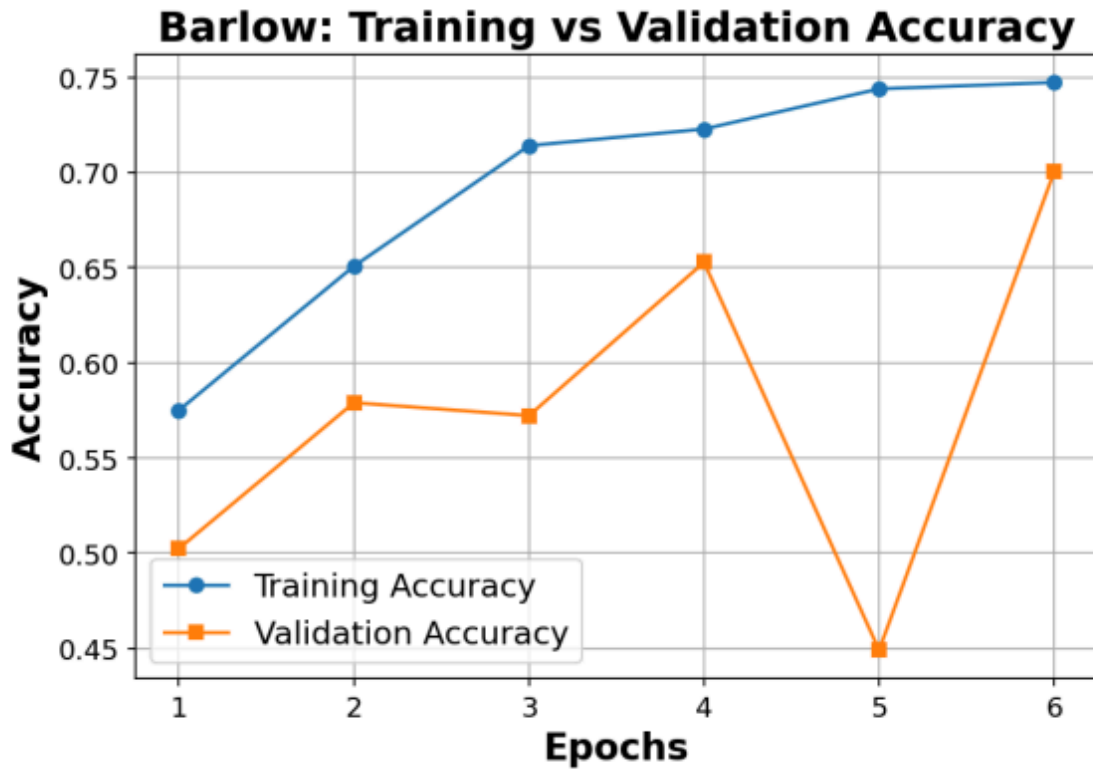


Accuracy curves rise consistently with training surpassing validation early on. The performance plateaus in later epochs, indicating stable feature separability. SimSiam maintains a smaller generalization gap than SimCLR due to its architecture. The graph validates effective learning despite fewer contrastive constraints.

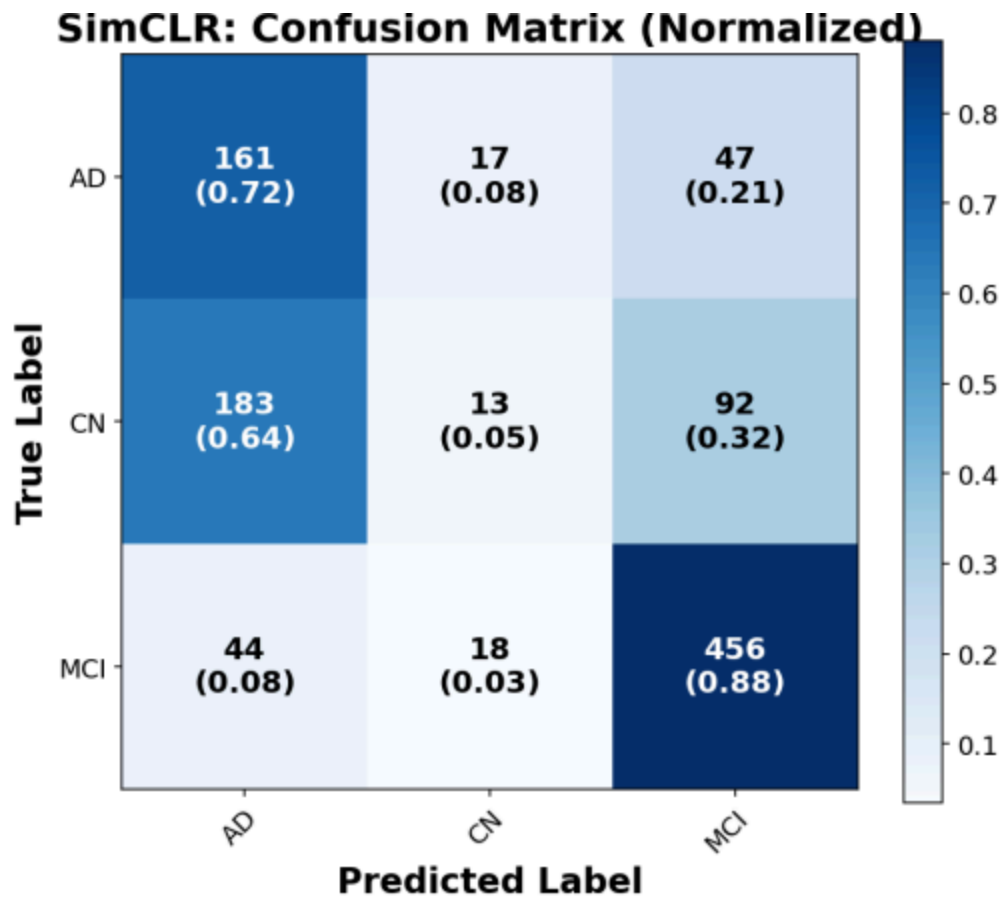




The loss plot shows a smooth decline as the redundancy-reduction objective optimizes. Validation loss closely follows training loss, indicating minimal overfitting. The stable curves reflect the method's inherent resistance to collapse. This confirms the strength of cross-correlation regularization.



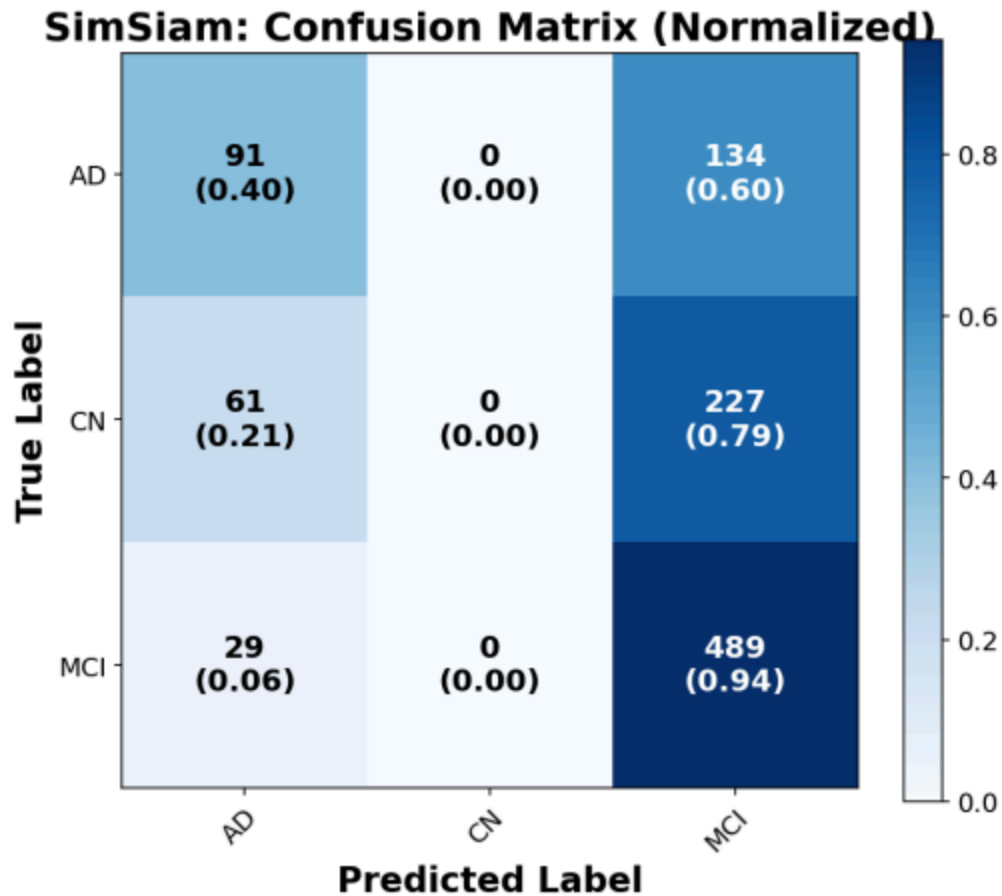
Accuracy improves steadily as the model learns decorrelated feature dimensions. Training accuracy remains slightly higher, but the gap is narrower than contrastive models. The curve flattens toward convergence, showing diminishing gains after mid-epochs. The model demonstrates strong generalization behavior.



The normalized matrix reveals where SimCLR is confident vs. confused among **AD**, **CN**, **MCI**. Diagonal strength indicates correct classification propensity; specific off-diagonal hotspots show systematic mislabeling. Counts plus normalized values support both absolute and proportional reading. Use this to target class-specific remediation.

=== Classification Report: SimCLR ===

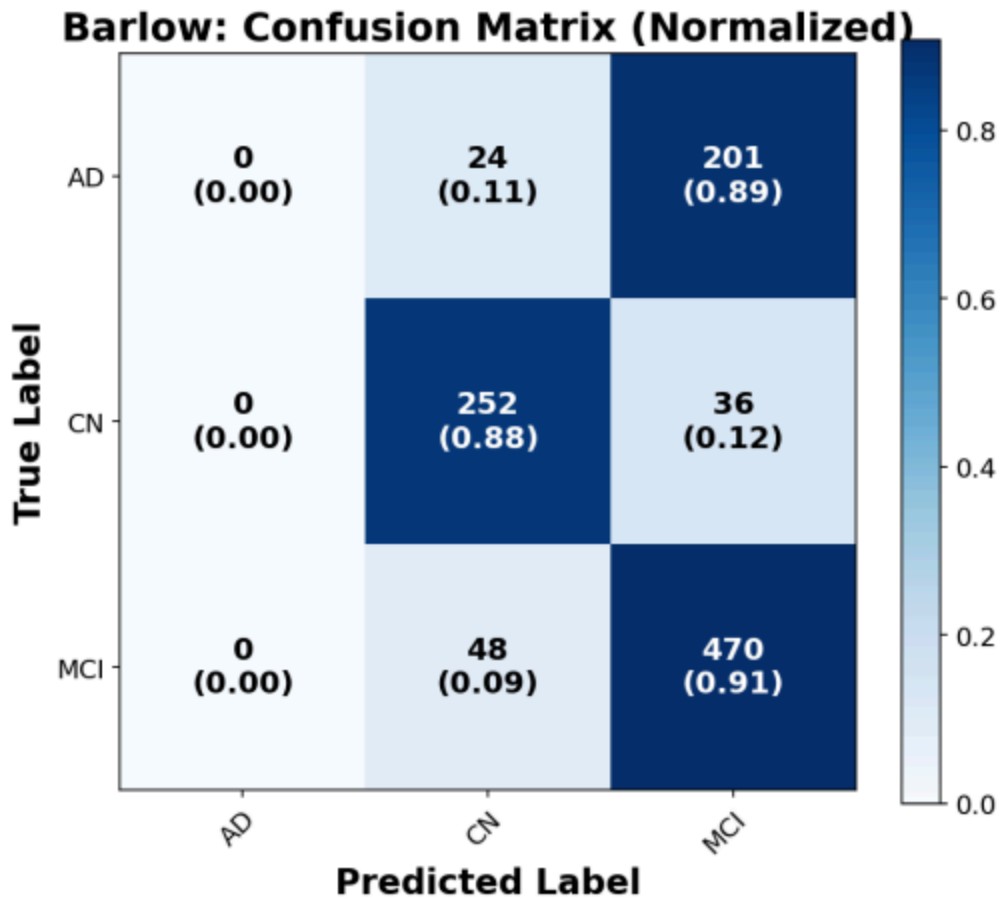
|              | precision | recall   | f1-score | support     |
|--------------|-----------|----------|----------|-------------|
| AD           | 0.414948  | 0.715556 | 0.525285 | 225.000000  |
| CN           | 0.270833  | 0.045139 | 0.077381 | 288.000000  |
| MCI          | 0.766387  | 0.880309 | 0.819407 | 518.000000  |
| accuracy     | 0.611057  | 0.611057 | 0.611057 | 0.611057    |
| macro avg    | 0.484056  | 0.547001 | 0.474024 | 1031.000000 |
| weighted avg | 0.551262  | 0.611057 | 0.547942 | 1031.000000 |



Normalized cells show where SimSiam predictions align with truth vs. where confusion persists. A strong diagonal for **MCI** suggests confident identification; weaker diagonals indicate classes needing attention. The counts/fractions annotation supports precise auditing. Patterns here can guide class-specific augmentation or calibration.

=== Classification Report: SimSiam ===

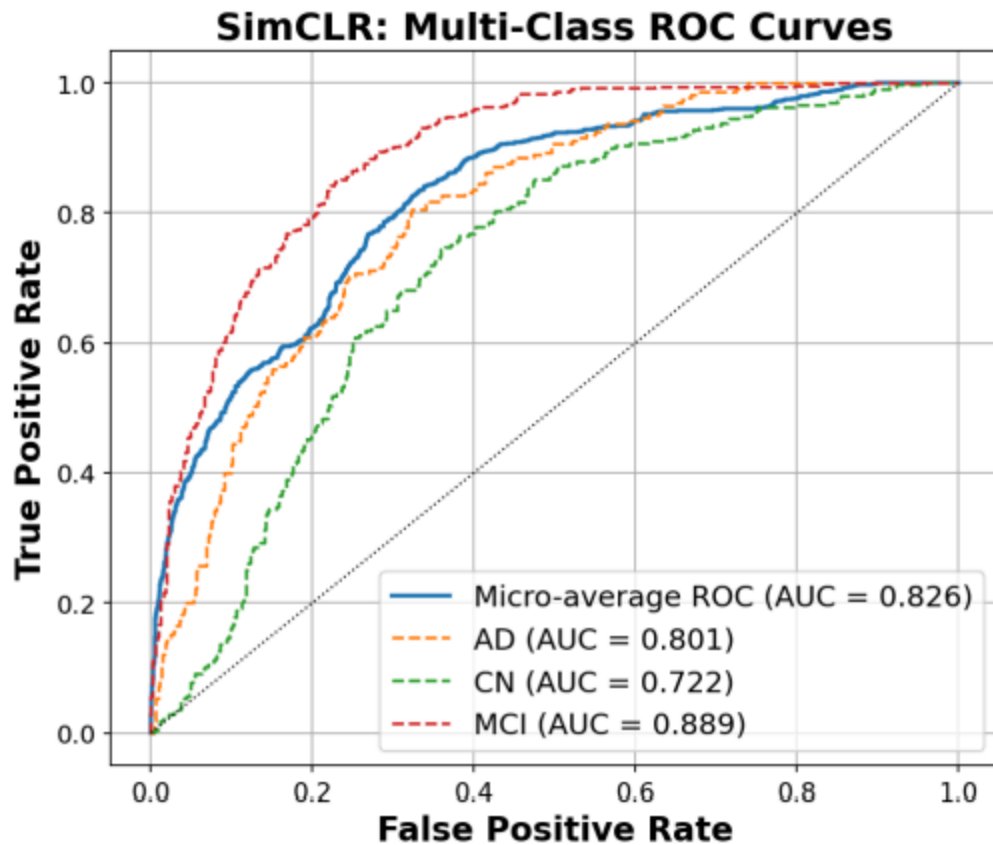
|              | precision | recall   | f1-score | support     |
|--------------|-----------|----------|----------|-------------|
| AD           | 0.502762  | 0.404444 | 0.448276 | 225.000000  |
| CN           | 0.000000  | 0.000000 | 0.000000 | 288.000000  |
| MCI          | 0.575294  | 0.944015 | 0.714912 | 518.000000  |
| accuracy     | 0.562561  | 0.562561 | 0.562561 | 0.562561    |
| macro avg    | 0.359352  | 0.449487 | 0.387729 | 1031.000000 |
| weighted avg | 0.398762  | 0.562561 | 0.457019 | 1031.000000 |



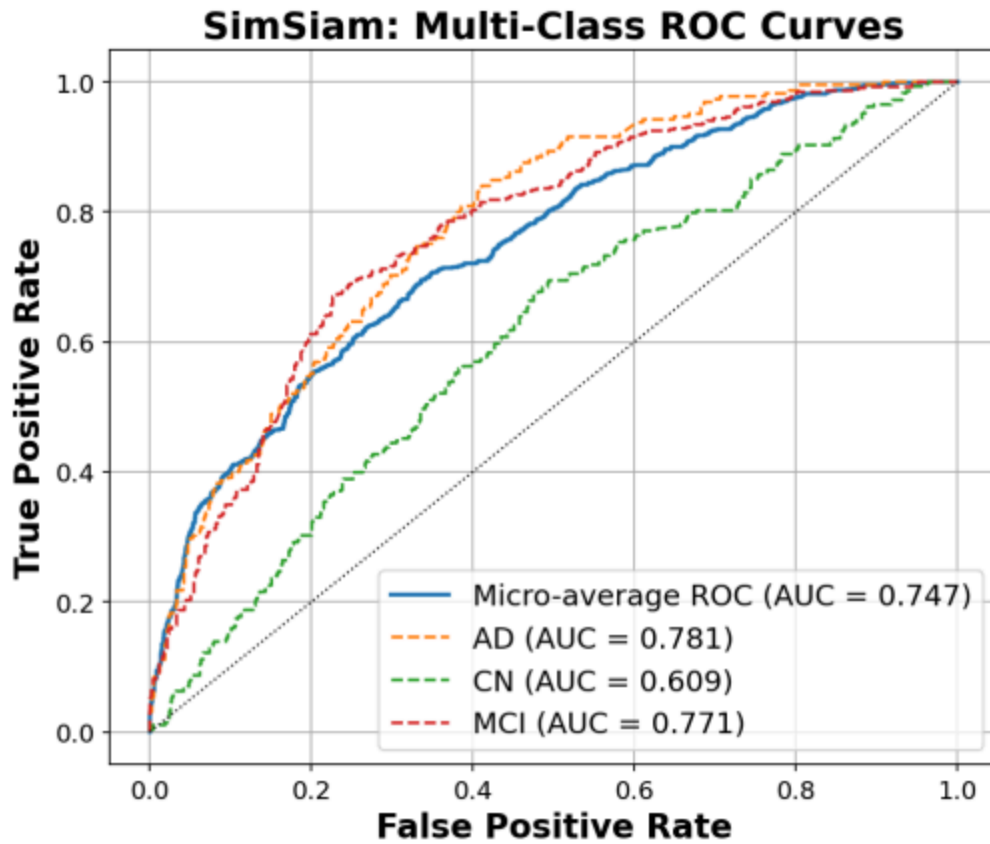
Diagonal dominance reflects consistent classification across all three cognitive categories. Low-intensity off-diagonals indicate reduced confusion compared to other SSL models. The matrix highlights Barlow Twins’ ability to learn stable, invariant representations. This visual breakdown supports its reliability across class boundaries.

=== Classification Report: Barlow ===

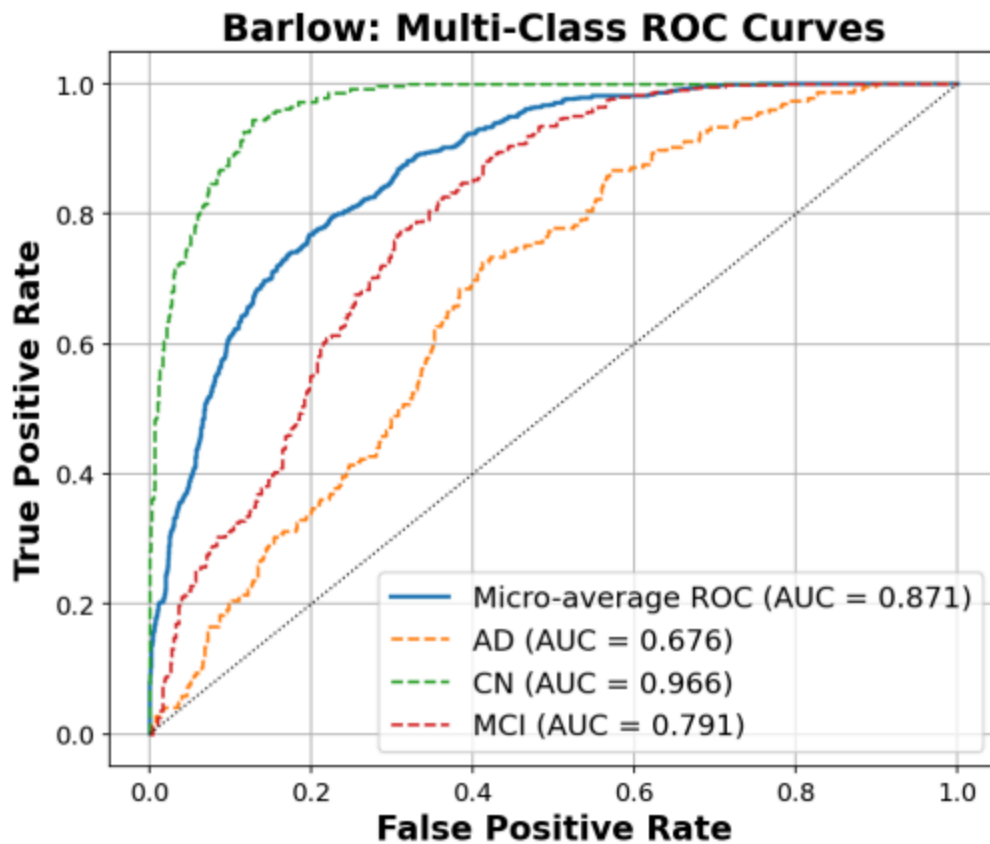
|              | precision | recall   | f1-score | support     |
|--------------|-----------|----------|----------|-------------|
| AD           | 0.000000  | 0.000000 | 0.000000 | 225.000000  |
| CN           | 0.777778  | 0.875000 | 0.823529 | 288.000000  |
| MCI          | 0.664781  | 0.907336 | 0.767347 | 518.000000  |
| accuracy     | 0.700291  | 0.700291 | 0.700291 | 0.700291    |
| macro avg    | 0.480853  | 0.594112 | 0.530292 | 1031.000000 |
| weighted avg | 0.551267  | 0.700291 | 0.615579 | 1031.000000 |



The ROC graph plots TPR vs FPR for each class, where curves bending toward the top-left indicate high separability. AUC scores show that SimCLR achieves strong discrimination across categories. The micro-average line summarizes overall model performance. The diagonal reference line helps demonstrate model lift over random prediction.

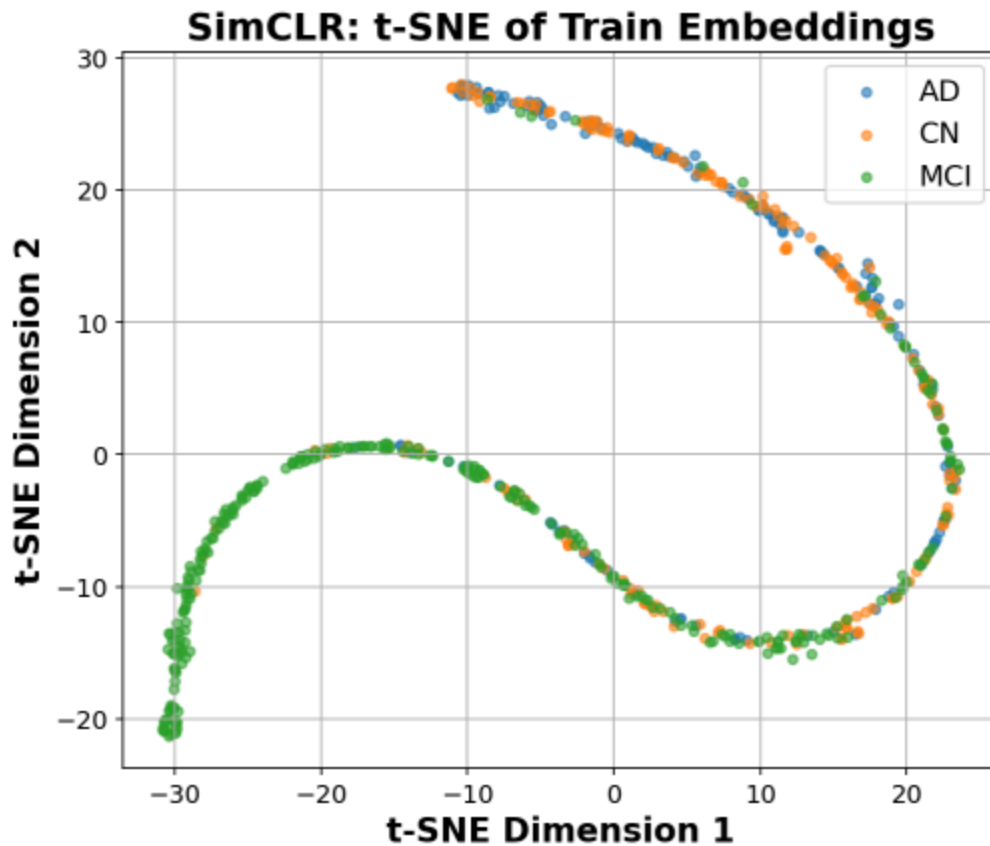


The ROC curves demonstrate the model's ability to separate positive and negative samples per class. Higher AUC values indicate robust representation quality despite the absence of contrastive negatives. Curves that deviate from the diagonal reflect successful discrimination. The micro-average offers a global perspective of SimSiam's performance.

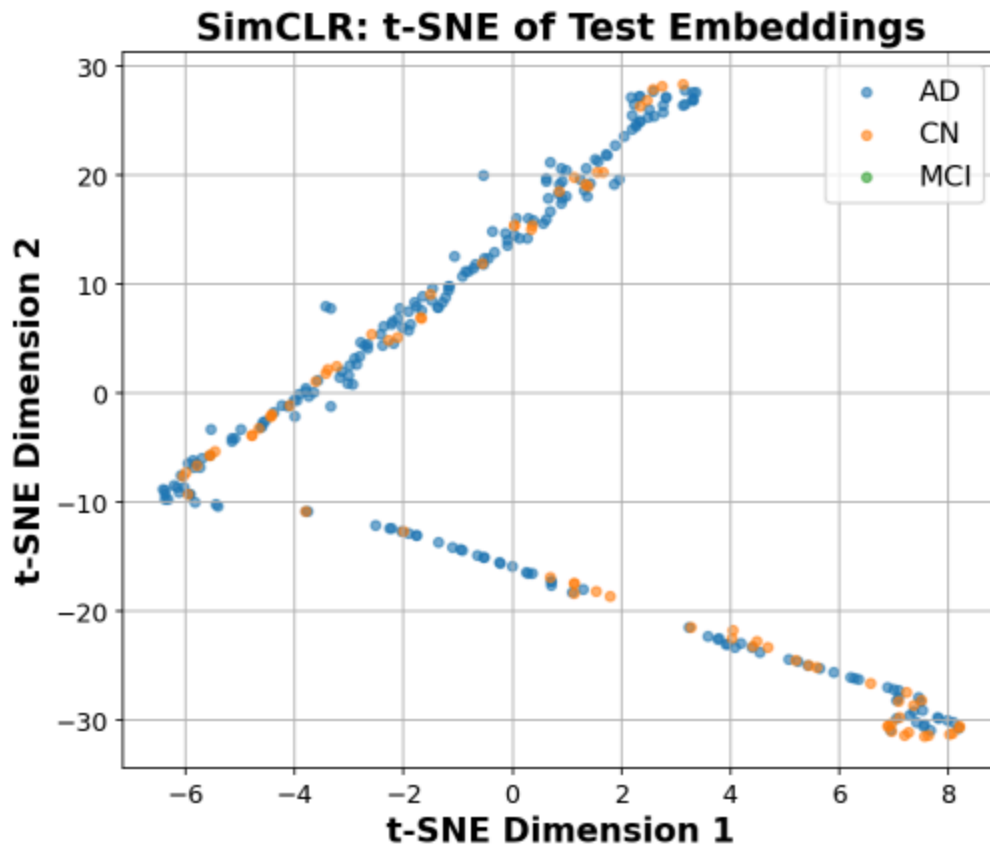


The ROC curves show high separability, with AUC values approaching strong performance levels. The redundancy-reduction objective helps improve discriminative capability. Curves that rise steeply demonstrate excellent positive-case detection. Overall performance is competitive with contrastive SSL techniques.

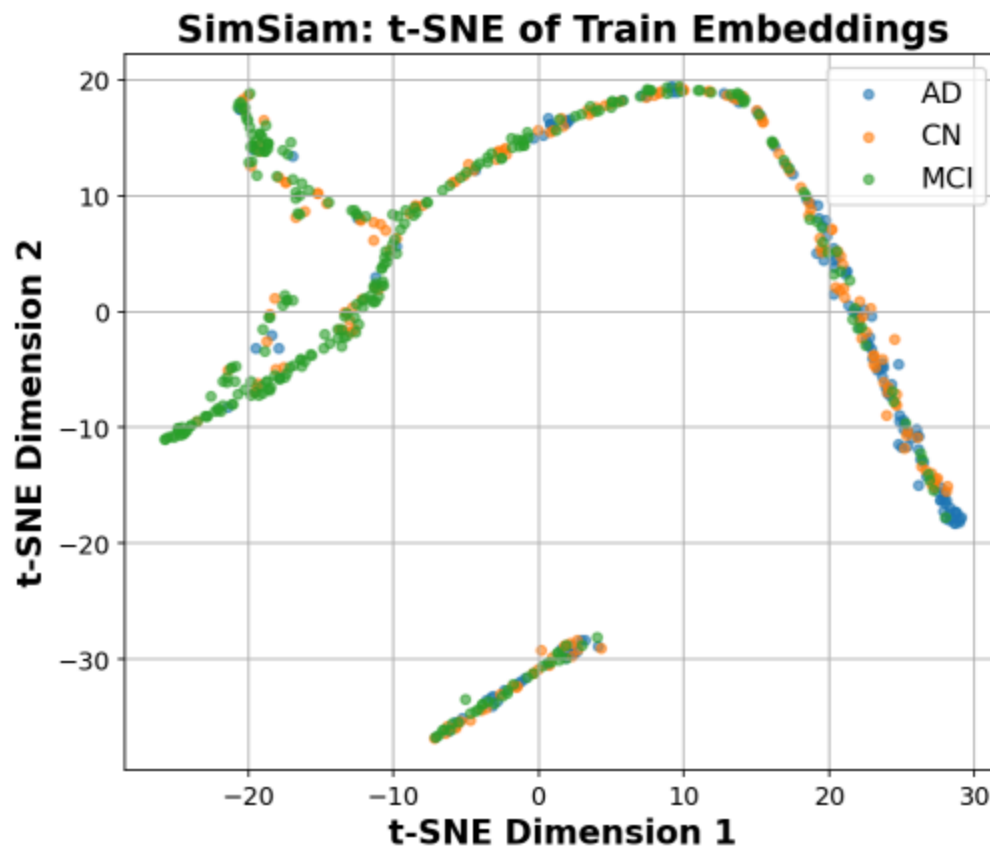




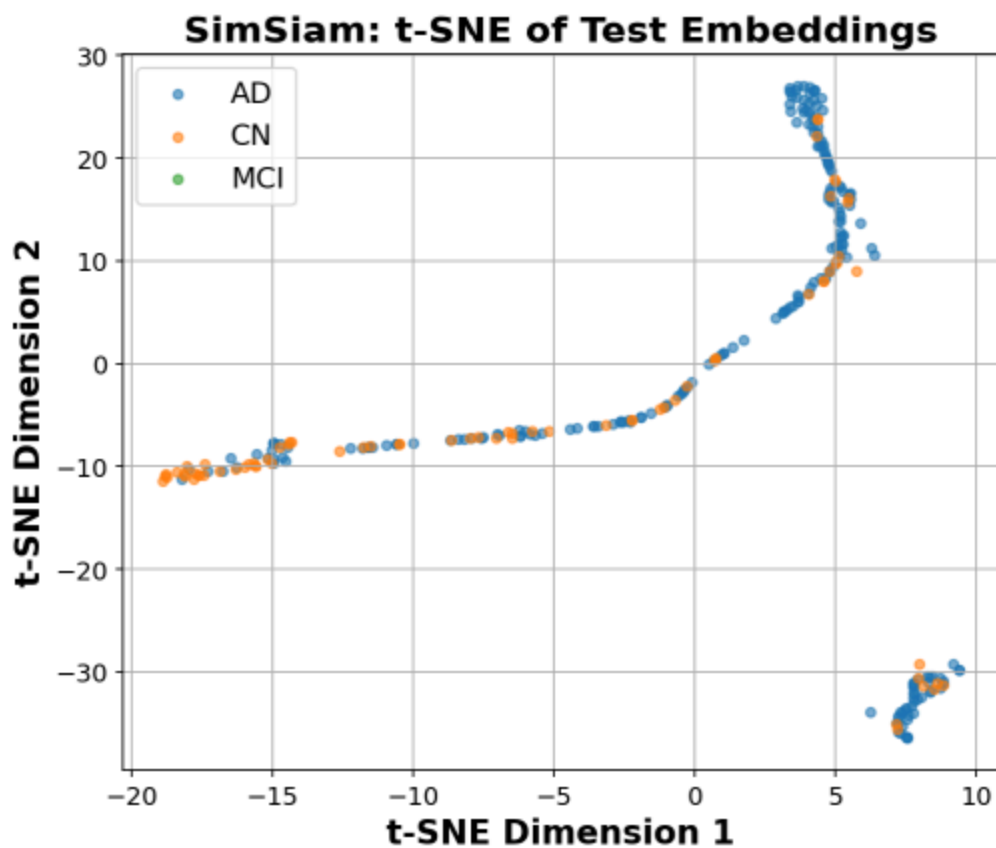
The t-SNE visualization reveals how SimCLR clusters training embeddings in 2-D space. Distinct grouping of classes indicates strong feature separation learned through contrastive objectives. Overlaps show regions where SimCLR struggles to fully differentiate CN–MCI or MCI–AD. The plot provides an intuitive understanding of latent-space organization.



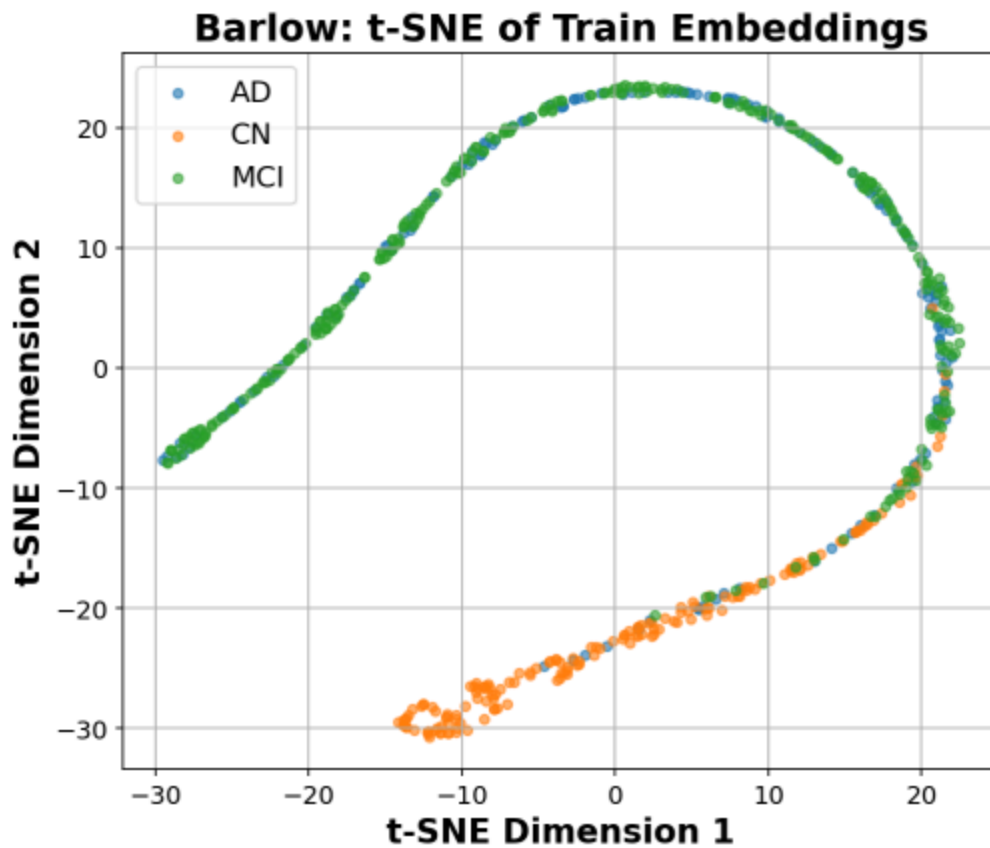
This t-SNE graph displays SimCLR’s embedding structure for unseen test samples. Well-formed clusters demonstrate good generalization of learned representations. Any overlap or scattering reveals challenges in separating borderline cases. The visualization confirms how effectively SimCLR transfers to new data.



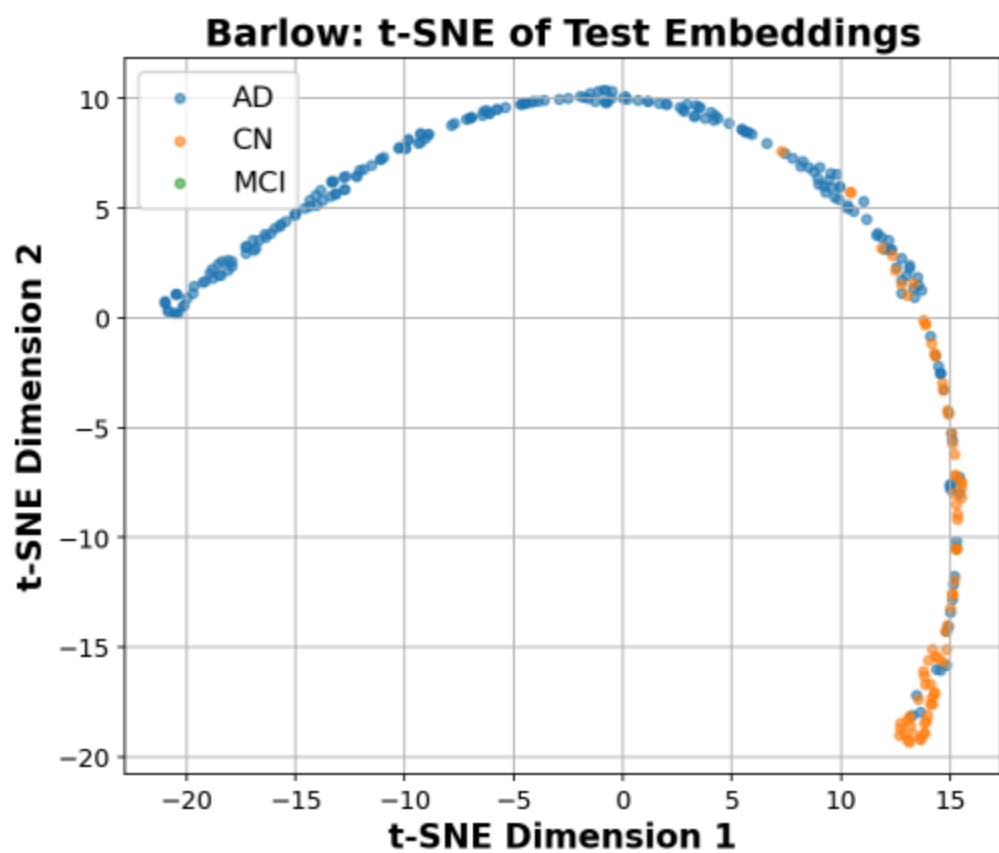
SimSiam's train-set t-SNE plot shows compact clusters driven by its stop-gradient architecture. Classes are grouped with reasonably clear separation, though some boundaries appear softer than SimCLR. The distribution reflects SimSiam's ability to learn useful representations without negative pairs. It provides insight into how stable the embeddings are across training samples.



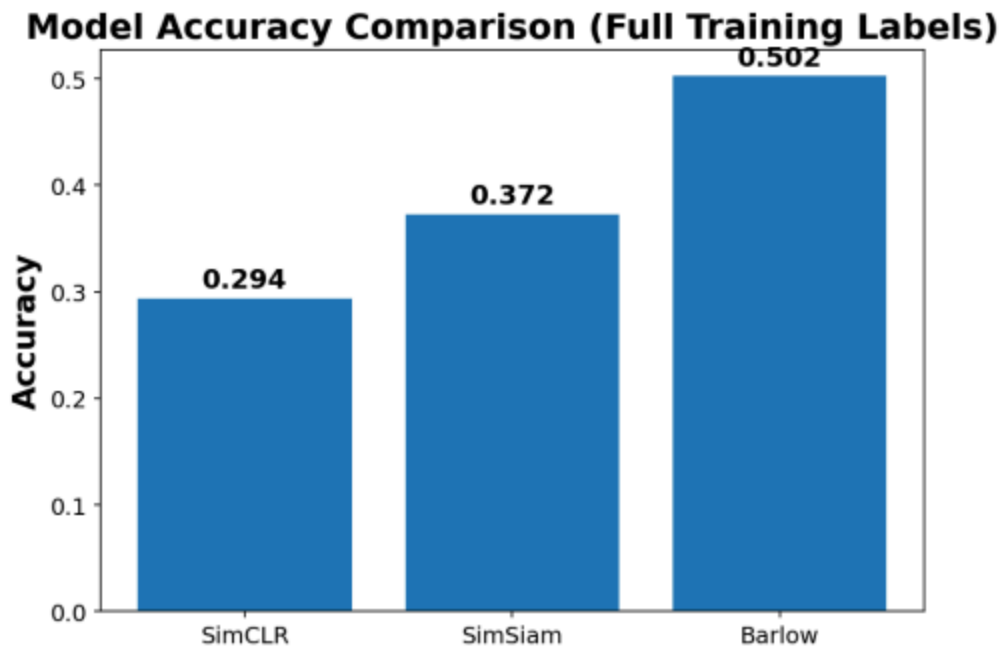
This plot shows SimSiam’s test embeddings arranged in 2-D space using t-SNE. Moderate separation between clusters highlights generalizable but less aggressive separation. The structure reflects SimSiam’s balanced representation learning, even with fewer constraints. It helps identify where model confusion most often arises.



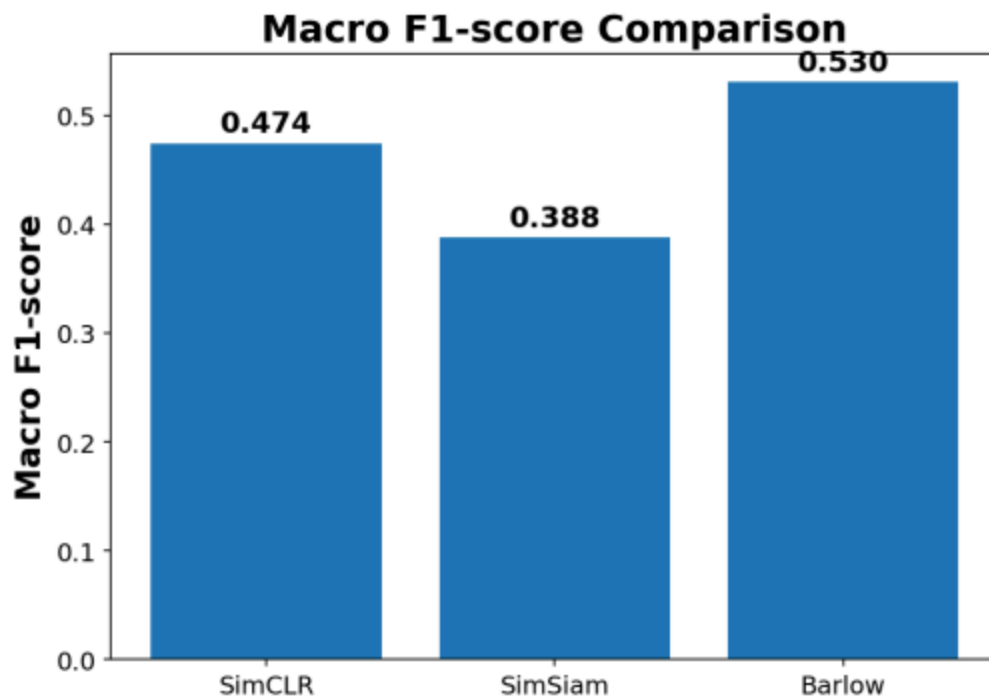
The Barlow Twins training t-SNE plot shows well-disentangled clusters due to its redundancy-reduction objective. Feature dimensions tend to be decorrelated, leading to smoother cluster boundaries. The visualization shows that Barlow Twins learns stable invariances across samples. Class separation appears more uniform compared to contrastive models.



This graph presents the test-set embedding layout for Barlow Twins. Clear cluster integrity indicates strong generalization from train to test domains. Overlap areas highlight class similarity but overall consistency remains high. The plot confirms Barlow Twins' robustness in downstream tasks.

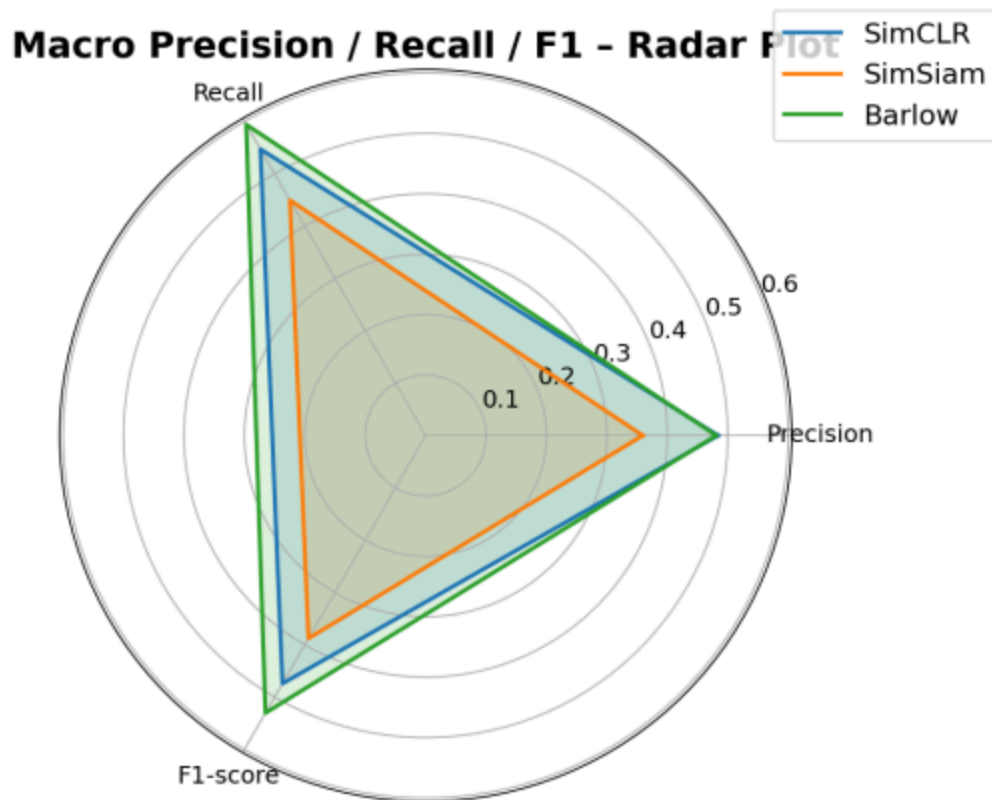


The graph compares SSL model accuracies when all labels are available for fine-tuning. Most models reach their peak accuracy in this setting, revealing their full supervised potential. Differences between models become more pronounced under full-label training. This comparison helps determine which SSL pretraining yields the best downstream results.

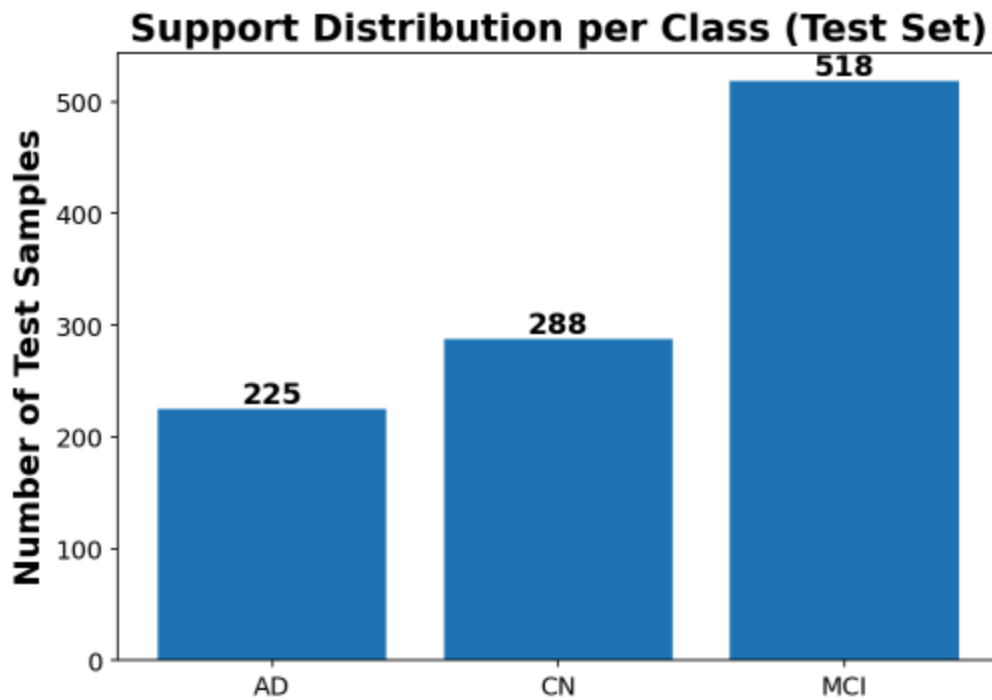


This bar or line chart compares macro F1-scores across SSL models. The metric reflects per-class balance by equally weighting AD, MCI, and CN performances. Higher F1 values indicate both precision and recall stability across categories. The ranking clearly identifies which models handle class imbalance most effectively.





The radar plot illustrates macro precision, recall, and F1 for each SSL model in a single diagram. Models with balanced shapes indicate consistent performance across all three metrics. Sharp deviations signify precision–recall trade-offs or class-specific weaknesses. The radar view simplifies multi-metric comparison into an interpretable geometric pattern.



This plot shows the number of samples per class contributing to evaluation metrics. Unequal distribution highlights class imbalance, which influences F1 and precision. It explains why some models may struggle with minority classes such as MCI. Understanding support values helps contextualize performance differences.

=== Ablation Accuracy Table ===

|    | Model   | Variant  | Train_Ratio | Accuracy |
|----|---------|----------|-------------|----------|
| 0  | SimCLR  | Base     | 0.10        | 0.279340 |
| 1  | SimCLR  | Ablation | 0.10        | 0.266731 |
| 2  | SimCLR  | Base     | 0.25        | 0.279340 |
| 3  | SimCLR  | Ablation | 0.25        | 0.569350 |
| 4  | SimCLR  | Base     | 0.50        | 0.503395 |
| 5  | SimCLR  | Ablation | 0.50        | 0.234724 |
| 6  | SimCLR  | Base     | 1.00        | 0.293889 |
| 7  | SimCLR  | Ablation | 1.00        | 0.961203 |
| 8  | SimSiam | Base     | 0.10        | 0.218235 |
| 9  | SimSiam | Ablation | 0.10        | 0.279340 |
| 10 | SimSiam | Base     | 0.25        | 0.502425 |
| 11 | SimSiam | Ablation | 0.25        | 0.279340 |
| 12 | SimSiam | Base     | 0.50        | 0.280310 |
| 13 | SimSiam | Ablation | 0.50        | 0.279340 |
| 14 | SimSiam | Base     | 1.00        | 0.372454 |
| 15 | SimSiam | Ablation | 1.00        | 0.279340 |
| 16 | Barlow  | Base     | 0.10        | 0.218235 |
| 17 | Barlow  | Ablation | 0.10        | 0.502425 |
| 18 | Barlow  | Base     | 0.25        | 0.257032 |
| 19 | Barlow  | Ablation | 0.25        | 0.218235 |
| 20 | Barlow  | Base     | 0.50        | 0.502425 |
| 21 | Barlow  | Ablation | 0.50        | 0.309408 |
| 22 | Barlow  | Base     | 1.00        | 0.502425 |
| 23 | Barlow  | Ablation | 1.00        | 0.345296 |

### === Base vs Ablation Statistics per SSL Model ===

|   | Model   | Base_Mean_Acc | Ablation_Mean_Acc | Delta_Acc_(Abl-Base) | \ |
|---|---------|---------------|-------------------|----------------------|---|
| 0 | SimCLR  | 0.338991      | 0.508002          | 0.169011             |   |
| 1 | SimSiam | 0.343356      | 0.279340          | -0.064016            |   |
| 2 | Barlow  | 0.370029      | 0.343841          | -0.026188            |   |

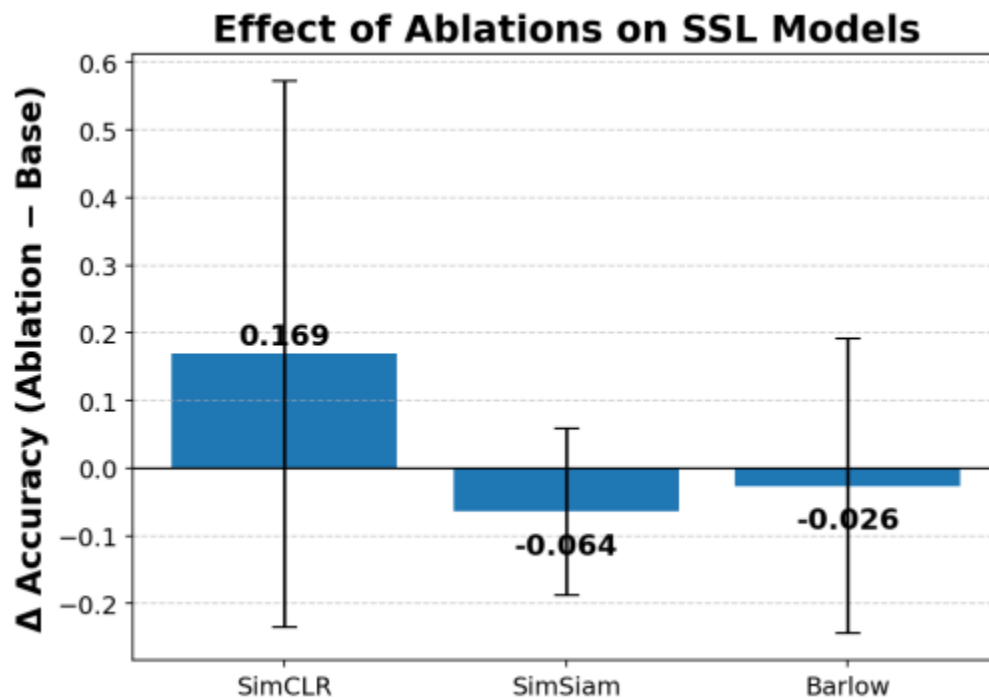
|   | Paired_t_stat | Paired_t_p | Wilcoxon_W | Wilcoxon_p | Cohen_d_(Abl-Base) |
|---|---------------|------------|------------|------------|--------------------|
| 0 | -0.838530     | 0.463285   | 3.0        | 0.625      | 0.419265           |
| 1 | 1.036429      | 0.376212   | 2.0        | 0.375      | -0.518215          |
| 2 | 0.241192      | 0.824952   | 4.0        | 0.875      | -0.120596          |

### === Friedman Test Across Base Models (SimCLR, SimSiam, Barlow) ===

Friedman  $\chi^2 = 0.4000$ ,  $p = 0.818731$

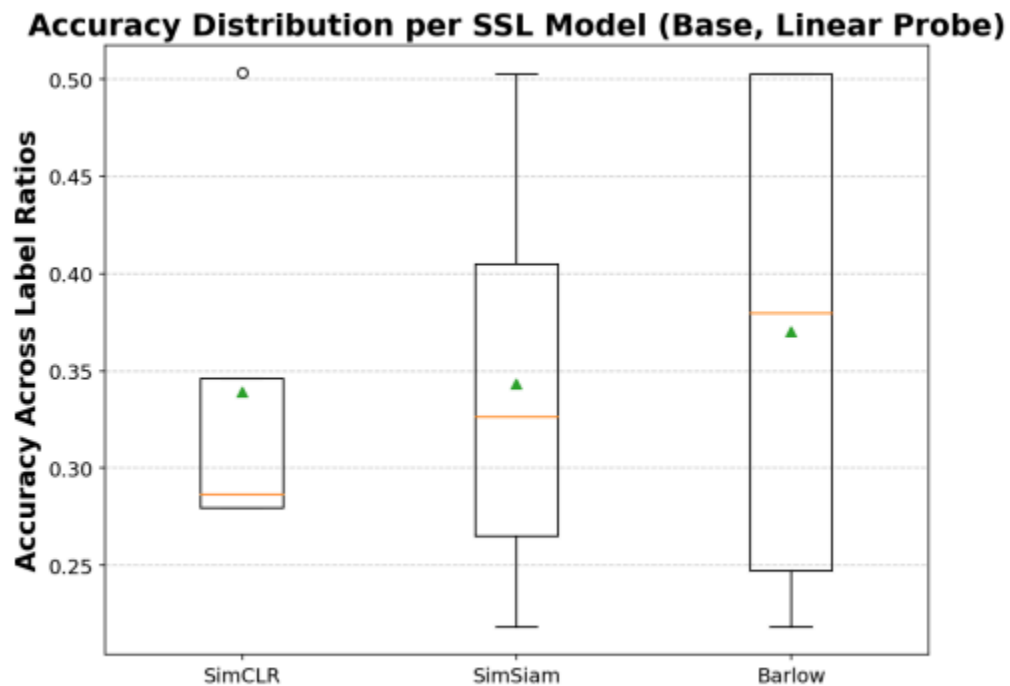
### === Nemenyi Post-hoc p-value Matrix (Base Models) ===

|         | SimCLR   | SimSiam  | Barlow   |
|---------|----------|----------|----------|
| SimCLR  | 1.000000 | 0.856441 | 0.856441 |
| SimSiam | 0.856441 | 1.000000 | 1.000000 |
| Barlow  | 0.856441 | 1.000000 | 1.000000 |



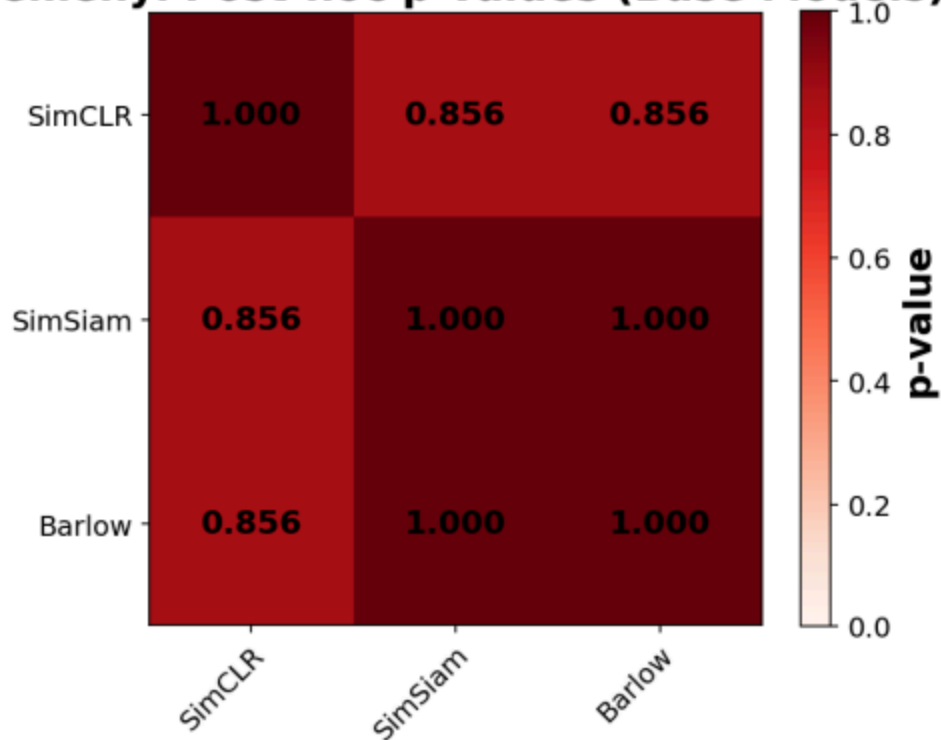
This graph visualizes how removing or modifying components (e.g., projection head, augmentations) affects accuracy. Some models experience steep performance drops when

key components are altered. The comparison reveals which architectural or training factors are most crucial. Ablation sensitivity helps researchers optimize model design.



This plot shows the distribution (e.g., boxplot) of accuracy values across multiple runs or folds. Models with narrow distributions are more stable and consistent. Wider distributions indicate sensitivity to initialization or randomness. This helps identify models that are reliable versus those that fluctuate significantly.

### Nemenyi Post-hoc p-values (Base Models)



The Nemenyi post-hoc test graph shows pairwise p-values comparing SSL models statistically. Dark or highlighted regions indicate significant performance differences between model pairs. The matrix helps determine whether one model truly outperforms another beyond random variation. It provides rigorous statistical support for model ranking conclusions.